

Estd: 1997

Eamcet Code: BVRI

BACHELOR OF TECHNOLOGY (B.Tech.) FOUR-YEAR DEGREE PROGRAM

**[Choice Based Credit System (CBCS)]
R22 Regulations**

(Applicable for the batches admitted from 2022-2023)

**DEPARTMENT OF CSE
(ARTIFICIAL INTELLIGENCE
AND MACHINE LEARNING)**



Department of CSE (AI & ML)

**B. V. RAJU INSTITUTE OF TECHNOLOGY
(UGC-Autonomous)**

**Approved by AICTE & Affiliated to JNTUH, Hyderabad
NAAC & NBA Accredited**

**Vishnupur, Narsapur, Medak Dist. – 502 313, Telangana, India.
(Sri Vishnu Educational Society)**

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DEPARTMENT OF CSE (AI & ML)
(B.TECH COURSE STRUCTURE APPLICABLE FOR STUDENTS ADMITTED FROM 2022-23)

I. STRUCTURE OF PROGRAM

The program consists of courses based on humanities and social sciences, basic sciences, engineering sciences, professional core, professional electives, open electives, industry internship, laboratories, mandatory courses and project work.

S. No.	Broad Course Classification	Course Group or Category	Course Description	Total Credits
1	Foundation Courses	Humanities, Social sciences & Management (HS)	English, humanities, social sciences, management	9.5
		Basic Sciences (BS)	Mathematics, Physics, Chemistry	24
		Engineering Sciences (ES)	Fundamental Engineering Courses	25
2	Core Courses and Core activities	Professional Core (PC)	Professional courses-core relevant to the discipline	54
		Project work (PW)	Minor & Major project (Institute/ Industry), Internship (Industry/ research organization), Seminar & technical term Paper (Institute)	14.5
3	Elective Courses	Professional Electives (PE)	Professional courses- electives relevant to the discipline	18
		Open Electives (OE)	Electives from other departments and or emerging areas	15
4	Mandatory Courses	Mandatory Courses (MC)	Courses on Environmental sciences, Ethical values, Indian constitution.	No Credits

II. INDUCTION PROGRAM

Induction Program (Mandatory)	3 Weeks duration
Induction program for students to be offered right at the start of the first year first semester	- Physical activity - Creative Arts - Universal Human Values - Literary - Proficiency Modules - Lectures by Eminent People - Visits to local Areas - Familiarization to Dept./Branch & Innovations

III. SEMESTER-WISE STRUCTURE OF CURRICULUM
B V RAJU INSTITUTE OF TECHNOLOGY
B.Tech. in CSE (Artificial Intelligence and Machine Learning)
COURSE STRUCTURE (R22 REGULATIONS)
Applicable for CSE(AI&ML), CSE(DS), AI&DS, BME, CHE, CIV, MECH, EEE
From A.Y 2022-23 admitted Student Batches

[L: Lecture, T: Tutorials, P: Practicals, C: Credits, CIE: Continuous Internal Examination, SEE: Semester End Examination]

I YEAR I SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1	A61NA	Linear Algebra and Calculus	BS	3	1		4	40	60	100
2	A61NB	Engineering Chemistry	BS	3	1		4	40	60	100
3	A61NC	English for Skill Enhancement	HS	3			3	40	60	100
4	A61HA	Computational Thinking and Prorgamming	ES	3			3	40	60	100
5	A611N	Engineering Chemistry Lab	BS			3	1.5	40	60	100
6	A612N	English Language and Communication Skills Lab	HS			3	1.5	40	60	100
7	A611H	Computational Thinking and Prorgamming Lab	ES			3	1.5	40	60	100
8	A612C	Engineering Prototyping Lab	ES			3	1.5	40	60	100
TOTAL							20	320	480	800

I YEAR II SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1	A62NA	Advanced Calculus	BS	3	1		4	40	60	100
2	A62NB	Applied Physics	BS	3	1		4	40	60	100
3	A62BB	Principles of Electrical and Electronics Engineering	ES	3			3	40	60	100
4	A62EA	Python Programming	ES	3			3	40	60	100
6	A621N	Applied Physics Lab	BS			3	1.5	40	60	100
7	A621B	Electrical and Electronics Lab	ES			3	1.5	40	60	100
8	A621E	Python Programming Lab	ES			3	1.5	40	60	100
9	A621C	Engineering Graphics Lab	ES			3	1.5	40	60	100
TOTAL							20	320	480	800

II YEAR I SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1	A63LA	Data Structures and Algorithms	PC	3			3	40	60	100
2	A63L1	Discrete Mathematics	PC	3			3	40	60	100
3	A63MA	Computer Organization	PC	3			3	40	60	100
4	A63E5	Database Management Systems	PC	3			3	40	60	100
5	A63NA	Applied Statistics and Numerical Methods	BS	3	1		4	40	60	100
6	A63EA	Introduction to Artificial Intelligence	ES	2			2	40	60	100
7	A631L	Data Structures and Algorithms Lab	PC			2	1	40	60	100
8	A643E	Database Management Systems Lab	PC			2	1	40	60	100
9	A634N	Environmental Science & Rural Sensitization	MC	2			-	-	-	-
TOTAL							20	320	480	800

II YEAR II SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1	A64KA	Operating Systems	PC	3			3	40	60	100
2	A64L1	Object Oriented Programming	PC	3			3	40	60	100
3	A64LA	Machine Learning	PC	3			3	40	60	100
4	A64L2	Automata and Compiler Design	PC	3			3	40	60	100
5		Open Elective-1	OE	3			3	40	60	100
6	A641L	Object Oriented Programming Lab	PC			2	1	40	60	100
7	A642L	Machine Learning Lab	PC			2	1	40	60	100
8	A641N	Business Communication Lab	HS			2	1	40	60	100
9	A641X	Minor Project	PW			4	2	40	60	100
10	A643N	Gender Sensitization	MC	2			-	-	-	-
TOTAL							20	360	540	900

III YEAR I SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1		Deep Learning	PC	3			3	40	60	100
2		Computer Networks	PC	3			3	40	60	100
3		Data Warehousing and Data Mining	PC	3			3	40	60	100
4		Fundamentals of Design Thinking	PC	3			3	40	60	100
5		Professional Elective-1	BS	3			3	40	60	100
6		Professional Elective-2	ES	3			3	40	60	100
7		Deep Learning Lab	PC			2	1	40	60	100
8		Data Warehousing and Data Mining Lab	PC			2	1	40	60	100
9		Constitution of India	MC	2			-	-	-	-
TOTAL							20	320	480	800

III YEAR II SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1		Data Analytics	PC	3			3	40	60	100
2		Natural Language Processing	PC	3			3	40	60	100
3		Business Economics and Financial Analysis	HS	3			3	40	60	100
4		Professional Elective-3	PE	3			3	40	60	100
5		Professional Elective-4	PE	3			3	40	60	100
6		Data Analytics Lab	PC			2	1	40	60	100
7		Natural Language Processing Lab	PC			2	1	40	60	100
8		Advanced Communication Skills Lab	HS			2	1	40	60	100
9		Mini Project Lab/Internship	PW			4	2	40	60	100
10		Intellectual Property Rights	MC	2			-	-	-	-
TOTAL							20	360	540	900

IV YEAR I SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1		PC 14	PC	3			3	40	60	100
2		PC 15	PC	3			3	40	60	100
3		PC 16	PC	3			3	40	60	100
4		Professional Elective-5	BS	3			3	40	60	100
5		Professional Elective-6	ES	3			3	40	60	100
6		PC Lab 9	PC			2	1	40	60	100
7		PC Lab 10	PC			2	1	40	60	100
8		Major Project – Phase I	PW			6	3	40	60	100
TOTAL							20	320	480	800

IV YEAR II SEMESTER

S.No	Course Code	Course Title	CAT	L	T	P	C	CIE	SEE	Total
1		Open Elective 2	OE	3			3	40	60	100
2		Open Elective 3	OE	3			3	40	60	100
3		Open Elective 4	OE	3			3	40	60	100
4		Seminar	PW			4	2	100		100
5		Major Project – Phase II	PW			18	9	40	60	100
	TOTAL						20	260	240	500

Professional Electives:

Semester	PE	Course Code	Course
III-I	PE1		Soft Computing
			Geometry of Vision
			Automated Reasoning and Theorem Proving
			Statistical Analysis for Data Science
			Knowledge Representation and Reasoning
	PE2		Expert Systems
			Game Programming
			Computational Linguistics
			Predictive Analytics
			Blockchain Technologies
III-II	PE3		Ensemble Learning
			Video Analytics
			Applied Natural Language Processing
			Business Intelligence
			DevOps
	PE4		Reinforcement Learning
			Virtual and Augmented Reality
			Sequence Models in NLP
			Social Networks and Semantic Web
			Data Modelling and Design
IV-I	PE5		Human Computer Interaction
			Introduction to Robotic Process Automation
			Speech and Audio Processing
			Social Media Analytics
			Supply Chain Analytics
	PE6		Information Retrieval Systems
			Introduction to Medical Image Analysis
			Machine Translation
			NoSQL Databases
			Industrial and Medical IoT

Structure of the Program:

S. No.	Broad Course Classification	Course Group or Category	Course Description	Total Credits
1	Foundation Courses	Humanities, Social Sciences & Management (HS)	English, Humanities, Social Sciences, Management	9.5
		Basic Sciences (BS)	Mathematics, Physics, Chemistry, Life Sciences	23
		Engineering Sciences (ES)	Fundamental Engineering Courses	21.5
2	Core Courses and Core activities	Professional Core (PC)	Professional Courses - Core relevant to the discipline	58
		Project Work (PW)	Minor Project, Mini Project/ Internship (Institute/ Industry), Major Project, Seminar	18
3	Elective Courses	Professional Electives (PE)	Professional Courses- Electives relevant to the discipline	18
		Open Electives (OE)	Electives from other Departments and or Emerging areas	12
4	Mandatory Courses	Mandatory Courses (MC)	Courses on Environmental Science, Gender Sensitization, Constitution of India, Intellectual Property Rights	No Credits
Total Credits				160

II YEAR I SEM
DATA STRUCTURES AND ALGORITHMS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A63LA	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Computational Thinking and Programming, Python Programming

Objectives: The Students will learn the following:

1. Understand the concept of algorithms and their characteristics, as well as the importance of analyzing their performance in terms of time and space complexity.
2. Understand the basic searching and sorting algorithms, and their implementation using a high-level programming language.
3. Understand hashing and the different collision resolution techniques, and how to implement them using arrays or linked lists.
4. Understand the concept of stacks and queues, and how to implement them using arrays or linked lists.
5. Understand linked lists, including single, double, and circular linked lists, and how to implement them.

Unit I:

(12 Lectures)

Preliminaries of Algorithms: Algorithm, Characteristics, Performance Analysis – Time and Space Complexity, Asymptotic Notations, Recursion – Recursive solutions for Factorial, GCD, Fibonacci, Towers of Hanoi.

Searching and Sorting: Linear Search and Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort. Time complexities computation for all searching and sorting techniques (Best, Average and Worst Cases)

Hashing: Representation, Collision Resolution Techniques - separate chaining, open addressing, linear probing, quadratic probing, double hashing and rehashing.

Unit II:

(12 Lectures)

Stacks: Basic Operations, Implementation using Array, Stack Applications: Infix to postfix Conversion, Evaluating Postfix Expressions.

Queues: Basic Operations, Implementation using Arrays. Circular Queue.

Linked Lists: Introduction, Operations on Single Linked List (SLL), Implementation of SLL, Double Linked List, Circular Linked List.

Unit III:

(10 Lectures)

Trees: Introduction, Binary Tree - Representations of Binary Tree, Binary Tree Traversals, Binary Search Tree, AVL Trees.

Graphs: Introduction, Representations – Array and Linked, Traversals - BFS, DFS.

Unit IV:

(08 Lectures)

Greedy Method: Introduction, Applications – Job Sequencing with deadlines, Fractional knapsack problem, Single Source Shortest Path, Minimum Cost Spanning Tree.

Unit V:

(10 Lectures)

Dynamic Programming: Introduction, Applications – Longest Common Subsequence, Travelling Salesman Problem, 0/1 knapsack, All Pairs Shortest Path, Reliability Design.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Analyze the time and space complexity of algorithms using asymptotic notations and perform recursive solutions for problems.
2. Implement and analyze various searching and sorting algorithms such as linear search, binary search, bubble sort, selection sort, insertion sort, and quicksort.
3. Implement data structures such as hashing, stacks, queues, linked lists, binary trees, and graphs.
4. Apply greedy method, dynamic programming, backtracking, and branch and bound techniques to solve various problems such as matrix multiplication, job sequencing, knapsack problems, shortest path, and graph problems.
5. Design efficient algorithms for solving real-world problems by selecting appropriate data structures and techniques, and analyze their time and space complexity

Textbook:

1. Data Structures, Second Edition, Richard F Gilberg, Forouzan, Cengage.

2. Fundamentals of Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekharan, Galgotia Publications Pvt. Ltd.

References:

1. Data Structures and Algorithms Made Easy, Narasimha Karumanchi.
2. Data Structures using C, Aaron M. Tanenbaum, Yedidyah Langsam, Moshe J Augenstein, Pearson.
3. Data structures and Algorithm Analysis in C, Mark Allen Weiss, Pearson Education. Ltd., Second Edition.
4. Design and Analysis of Algorithms, Aho, Ullman and Hopcroft, Pearson Education.

DISCRETE MATHEMATICS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A63L1	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Nil

Objectives: The Students will learn the following:

- To introduce the concepts of mathematical logic.
- To perform the operations associated with sets, relations and functions.
- To learn the types of algebraic structures and their properties
- To learn methods for solving counting problems.
- To use Graph Theory concepts for solving real time problems

Unit I:

(12 Lectures)

Mathematical Logic: Introduction, Statements and Notation, Connectives, Normal Forms, Theory of Inference for the Statement Calculus, The Predicate Calculus, Inference Theory of the Predicate Calculus.

Unit II:

(10 Lectures)

Sets and Relations: Introduction, Basic Concepts of Set Theory, Cartesian Products, Representation of Discrete Structures, Binary Relation- Properties of Binary Relation, Partition and Covering of a set, Equivalence Relation, Compatibility Relation, Partial Ordering Relation, Hasse Diagram, Functions- definition, properties, One-to-one functions, Onto functions, Bijective functions, Inverse and Composite Functions, Transitive Closure.

Unit III:

(11 Lectures)

Algebraic Structures: Introduction, Binary operation, properties of binary operation, Algebraic Systems, Semigroups and Monoids, Groups, Subgroups, and Homomorphism, Cyclic group, Lattice and its properties.

Boolean Algebra: Identities of Boolean Algebra, Duality, Representation of Boolean Function, Boolean Ring.

Unit IV:

(10 Lectures)

Elementary Combinatorics: Basics of Counting, Combinations and Permutations, Enumeration of Combinations and Permutations, Enumerating Combinations and Permutations with Repetitions, Enumerating Permutation with Constrained Repetitions, Binomial Coefficient, The Binomial and Multinomial Theorems, The Principle of Inclusion-Exclusion, Pigeon Hole Principle.

Unit V:

(12 Lectures)

Graph Theory: Basic Concepts, Isomorphism and Subgraphs, Trees and their Properties, Spanning Trees, Directed Trees, Binary Trees, Planar Graphs, Euler's Formula, Multi-graphs and Euler Circuits, Hamiltonian Graphs, Chromatic Numbers, The Four-Color Problem.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply mathematical logic to find the logical equivalences and implications of statement formulas.
2. Describe the properties and types of sets, relations and functions.
3. Identify the properties and types of algebraic structures.
4. Solve counting problems by using concepts of combinatorics.
5. Apply graph theory concepts in solving real-world problems.

Textbooks:

1. Discrete Mathematical Structure and It's Application to Computer Science, J.P. Tremblay, R. Manohar, TMG Edition, Tata McGraw-Hill.
2. Discrete Mathematics for Computer Scientists & Mathematicians, Joe L. Mott, Abraham Kandel, Theodore P. Baker, Pearson , 2nd Edition.

References:

1. Discrete and Combinatorial Mathematics - An Applied Introduction, Ralph. P, Grimald, Pearson Education, 5th Edition
2. Discrete Mathematical Structures: Thomas Kosy, Tata McGraw Hill Publishing Co.
3. Discrete Mathematics and its Applications, Kenneth H. Rosen, Tata McGraw – Hill.
4. Elements of Discrete Mathematics A Computer Oriented Approach, C L Liu and D P Mohapatra, 3rd Edition, Tata McGraw – Hill.
5. Discrete Mathematics, Norman L. Biggs, Seymour Lipschutz, Marc Lipson, 2nd Edition, Oxford University Press. Schaum's Outlines Series.

COMPUTER ORGANIZATION

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A63MA	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites:

Objectives: The Students will learn the following:

- To learn the basic principles of computer organization and architectural design concepts.
- To learn Microprogrammed control of a simple digital computer and Register organization and different Instruction formats.
- To understand different types of Data Representations and Computer Arithmetic operations.
- To understand Input-Output organization and Memory organization.
- To learn the concepts of Pipeline and Multi Processors.

Unit I:**(10 Lectures)**

Data Representation: Data types, Complements, Fixed Point Representation, Floating Point Representation.

Computer Arithmetic: Addition and subtraction, Multiplication Algorithms, Division Algorithms, Floating – point Arithmetic operations.

Unit II:**(10 Lectures)**

Digital Computers: Introduction, Block diagram of Digital Computer, Definition of Computer Organization, Computer Design and Computer Architecture.

Register Transfer: Register Transfer language, Register Transfer, Bus and memory transfers.

Basic Computer Organization and Design: Instruction codes, Computer Registers, Computer Instructions, Timing and Control, Instruction cycle, Memory Reference Instructions, Input – Output and Interrupt.

Unit III:**(10 Lectures)**

Microprogrammed Control: Control memory, Address sequencing, Micro program example, Design of control unit.

Central Processing Unit: General Register Organization, Instruction Formats, Addressing modes, Data Transfer and Manipulation, Program Control.

Unit IV:**(10 Lectures)**

Input-Output Organization: Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupt, Direct memory Access.

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary memory, Cache Memory.

Unit V:**(10 Lectures)**

Reduced Instruction Set Computer: CISC Characteristics, RISC Characteristics.

Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processor.

Multi Processors: Characteristics of Multiprocessors, Interconnection Structures, Cache Coherence.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Understand basic Computer Instructions and their impact on processor design.
2. Demonstrate an understanding of Micro programmed control and Register Organization.
3. Recognize and manipulate representations of numbers stored in digital computers and perform different Arithmetic operations.
4. Understand in a better way the I/O and Memory Organization in detail.
5. Design a pipeline for consistent execution of instructions with minimum hazards and understand Multi Processor concepts.

Textbooks:

1. Computer System Architecture – M. Morris Mano, Third Edition, Pearson/PHI.

References:

1. Computer Organization – Carl Hamacher, Zvonks Vranesic, Safea Zaky, V th Edition, McGraw Hill.
2. Computer Organization and Architecture – William Stallings Sixth Edition, Pearson/PHI.
3. Structured Computer Organization – Andrew S. Tanenbaum, 4 th Edition, PHI/Pearson.

DATABASE MANAGEMENT SYSTEMS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A63E5	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Computational Thinking and Programming, Python Programming

Objectives: The Students will learn the following:

- To learn the fundamentals of database systems and analyze the data models to represent a database system using ER diagrams.
- To learn the Relational Model using SQL and formulate basic SQL Queries.
- To learn the Schema Refinement approaches and functional dependencies for effective Database Design.
- To Learn the Normalization rules for effective database design.
- To learn the Importance of Transactions, Concurrency Control mechanism to real-time environment and analyze the importance of recovery management
- To learn the internal storage structures using different file and indexing techniques which will help in physical Database design.

Unit I: (10 Lectures)

Introduction to Database System: Introduction-Database System Applications, Purpose of Database Systems, View of Data, Instance and Schema, Data Models, Database Languages - DDL, DML, DCL, TCL, Database System Structure, History of Data base Systems

Introduction to Database Design: Introduction to Data base design steps, ER diagram, Keys, Beyond ER Design, Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Conceptual Design for Large enterprises: Case study.

Unit II: (10 Lectures)

Introduction to Relational Model: Structure of Relational Model, Constraints, Relational Algebra, Relational Calculus: TRC, DRC.

Form of Basic SQL Query: SQL fundamentals, Examples of Basic SQL Queries, Introduction to Nested Queries, Correlated Nested Queries, Set - Comparison Operators, Aggregate Operators, NULL values, Logical connectives, Outer Joins, Complex Integrity Constraints in SQL, Views, Triggers.

Unit III: (12 Lectures)

Relational-Database Design: Problems Caused by redundancy, Decompositions - Problem related to decomposition, Functional Dependencies - Reasoning about FDS, Normal Forms - FIRST, SECOND, THIRD Normal forms - BCNF - Properties of Decompositions - Loss less join Decomposition, Dependency preserving Decomposition, Multi valued Dependencies - FOURTH Normal Form, Join Dependencies, FIFTH Normal form, Inclusion Dependencies.

Unit IV: (10 Lectures)

Transaction and Concurrency Control – ACID Properties, Schedules, Serializability, need for Concurrency, Locking Protocols – Two Phase Locking, Deadlock, Timestamp Based Protocols, Validation Based Protocols.

Recovery Systems: Failure Classification, Log Based Recovery, ARIES, Check Points, Save Point, Buffer Management, Remote Back Up Systems.

Unit V: (11 Lectures)

Overview of Storage and Indexing: Data on External Storage, overview of physical storage media, RAID, File Organization and Indexing, Data Dictionary Storage, Cluster Indexes, Primary and Secondary Indexes, Index data Structures – Hash Based Indexing, Tree base Indexing, B+ Trees: A Dynamic Index Structure.

Outcomes: Upon the successful completion of the course, the student will be able:

- Demonstrate the fundamental elements of database systems and database design.
- Identify the importance of relational model and solve Queries using SQL.
- Apply Functional dependencies and decomposition rules for database design.
- Identify the Transaction Processing systems and Recovery methods.
- Compare basic database storage structures and access techniques.

Textbooks:

- Database System Concepts, Abraham Silberschatz, Henry F. Korth, S. Sudharshan, 6th Edition, Tata McGraw Hill, 2011.
- Database Management Systems, Raghurama Krishnan, Johannes Gehrke, McGrawHill Education, 3rd Edition, 2003.

References:

1. An Introduction to Database Systems, C. J. Date, A.Kannan, S. Swamynathan, 8th Edition, Pearson Education, 2006.
2. Fundamentals of Database Systems, Ramez Elmasri, Shamkant B. Navathe, 6th Edition, Pearson, Education, 2011.
3. Database Management Systems, G. K. Gupta, Tata McGraw Hill, 2011.

APPLIED STATISTICS AND NUMERICAL METHODS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A63NA	BS	L	T	P	C	CIE	SEE	Total
				3	1	-	4	40	60	100

Pre-requisites: Linear Algebra and Calculus, Advanced Calculus

Objectives: The Students will learn the following:

1. To enable the students to understand the concepts of probability and Random Variables.
2. To understand Probability distributions, correlation and regression.
3. To create an understanding on various types of sampling distribution and estimations.
4. To train students the basic principles of Statistical Inference.
5. To find roots of a nonlinear equations, system of non-homogenous equation, concept of interpolation and solutions of Initial Value Problems (IVP) using various numerical methods.

Unit I**(08 Lectures)**

Random Variables: Review of probability, Moments, Random Variables: Discrete and Continuous - Probability distributions, mass function/ density function of a probability distribution and Distribution function. Mathematical Expectation, Moment generating function.

Unit II**(12 Lectures)**

Probability distributions: Binomial, Poisson and Normal Distribution. Evaluation of Statistical parameters: Mean Variance and their properties.

Correlation: Correlation-Coefficient of Correlation, The rank correlation.

Regression: The lines of regression, properties of regression coefficients.

Unit III**(08 Lectures)**

Sampling Distribution: Definitions of population, sample, parameter and statistic, types of sampling, sampling distribution, Standard error, and Sampling distribution of means and sampling distribution of variance. Estimation: Point and Interval estimations, Criteria of a good estimator, estimators for means and proportions.

Unit IV**(12 Lectures)**

Testing of Hypothesis: Null Hypothesis, Alternative Hypothesis, Type-I and Type-II Errors, critical region, confidence interval, level of significance, one-tail, two-tail tests, Maximum Error.

Large sample tests: Z-test for single mean, difference of means, single proportion and difference of proportions.

Small sample tests: Student t-test for single mean, difference of means and F-Test for equality of two population variances, Chi-square test for goodness of fit and independence of attributes.

Unit V**(10 Lectures)**

Solution of algebraic and transcendental equations: Bisection method, Newton-Raphson method.

Solution of system of non-homogeneous equations: Gauss Seidel iteration method.

Interpolation: Finite differences, Interpolation using Newton's forward and backward difference formulae.

Solving first order Ordinary differential equations: Taylor's series, Euler and modified Euler's methods, Runge - Kutta method of fourth order.

Outcomes: Upon the successful completion of the course, the student will be able:

1. To understand the probability in a certain realistic situation using different random variables and calculate expectations and moments.
2. Solve statistical problems using Binomial, Poisson, Normal distributions and also distinguish correlation and regression analysis and interpret them.
3. Calculation of sample mean, proportions and estimations of given statistical data.
4. Examine the given statistical hypothesis.
5. Solve algebraic equations, system of non-homogeneous equations using numerical techniques, interpolate the given experimental data and obtain approximate solutions for ODE.

Textbook:

1. S. Grewal, Higher Engineering Mathematics, 42nd Edition, Khanna Publishers
2. Probability and Statistics for Engineers, Miller, John E. Freund, PHI.

References:

1. Advanced Engineering Mathematics, R. K. Jain & S. R. K. Iyengar, 3rd Edition, Narosa Publishing House, Delhi.

2. Fundamentals of Mathematical Statistics, S. C. Gupta & V. K. Kapoor, Sultan Chand
3. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons Publisher.
4. Probability and Statistics for Engineers, Miller, John E. Freund, PHI
5. Higher Engineering Mathematics, B.V. Ramana, Tata Mc Grawhill.
6. Introductory Methods of Numerical Analysis, S.S. Sastry, PHI.

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A63EA	ES	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Mathematics

Objectives: The Students will learn the following:

1. State Space Searching techniques to solve a real-world problem
2. Heuristic Searching techniques to optimize a problem iteratively
3. Randomized Search and Emergent Systems to determine the best solution for the developed model
4. Optimal Path algorithms to determine shortest paths in different circuits
5. Problem Decomposition to decompose a problem into manageable sub-components is a necessity in complex problem-solving tasks

Unit I**(09 Lectures)**

Introduction: Artificial Intelligence, Historical Backdrop, What is Artificial Intelligence, AI Agents –types of Agents, Applications of AI.

State Space Search: Generate and Test, Simple Search, Depth First Search (DFS), Breadth First Search (BFS), Comparison of BFS and DFS, Depth Bounded DFS (DBDFS), Depth First Iterative Deepening (DFID).

Unit II**(10 Lectures)**

Heuristic Search: Heuristic Functions, Best First Search, Hill Climbing, Local Maxima, Solution Space Search, Variable Neighborhood Descent, Beam Search, Tabu Search, Peak to Peak Methods.

Unit III**(10 Lectures)**

Randomized Search and Emergent Systems: Iterated Hill Climbing, Simulated Annealing, Genetic Algorithms, the Travelling Salesman Problem, Neural Networks, Emergent Systems.

Unit IV**(10 Lectures)**

Finding Optimal Paths: Brute Force, Branch and Bound, Refinement Search, Dijkstra's Algorithm, Algorithm A*, Admissibility of A*, Iterative Deepening A* (IDA*), Recursive Best First Search (RBFS), Pruning the Closed List, Pruning the Open List, Divide and Conquer Beam Stack Search.

Unit V**(09 Lectures)**

Problem Decomposition: SAINT, Dendral, Goal Trees, Rule based Systems, XCON, Rule Based Expert Systems.

Case Studies: Autonomous Vehicle, Medical Image Analysis.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply appropriate State Space Searching techniques to solve a real-world problem.
2. Apply appropriate Heuristic Searching techniques to solve a real-world problem.
3. Analyze the problem and infer new knowledge using Randomized Search and Emergent Systems.
4. Apply Optimal Paths Algorithms to real world problems.
5. Analyze the Problem Decomposition Methods and their test cases.

Textbook:

1. A First Course in Artificial Intelligence, Deepak Khemani, Tata Mc-Graw Hill Education, 2013.

References:

1. AI – A Modern Approach, Stuart Russel, Peter Norvig, 4th Edition, Pearson Education, 2010.
2. Artificial Intelligence, Kevin Knight, Elaine Rich, Shivshankar B. Nair, 3rd Edition, Tata Mc-Graw Hill Education, 2019.

DATA STRUCTURES AND ALGORITHMS LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A631L	PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites:

Objectives: The Students will learn the following:

Week 1:

- Write a program to find GCD of two given integers using recursive function.
- Write a program to find Nth Fibonacci Number using recursive function.
- Write a program for Towers of Hanoi using recursion: N disks are to be transferred from Peg S to Peg D with Peg I as the intermediate Peg.

Week 2:

- Write a program to find an element in given list of elements using Linear Search.
- Write a program to find an element in given list of elements using Binary Search

Week 3:

- Write a program that uses a function to implement Bubble sort.
- Write a program that uses a function to implement selection sort.
- Write a program that uses a function to implement Insertion sort.
- Write a program that uses a function to implement Quick sort.

Week 4:

- Write a program that implement Hashing using Linear Probing.
- Write a program that implement Hashing using Quadratic Probing.

Week 5:

- Write a program that implement stack (its operations) using arrays.
- Write a program that uses Stack operations to convert infix expression into postfix expression.
- Write a program that uses Stack operations to evaluate a postfix expression.

Week 6:

- Write a program that implement Queue (its operations) using arrays.
- Write a program that implement Circular Queue (its operations) using arrays.

Week 7:

- Write a program that uses functions to create a singly linked list and its operations (insert, delete, search).
- Write a program to Create a Double Linked list and its operations (insert, delete, search).

Week 8:

- Write a program to implement STACK using Linked List
- Write a program to implement QUEUE using Circular Linked List.

Week 9:

- Write a program to create a Binary Search Tree and its operations.
- Write a recursive program, for traversing a binary tree in pre-order, in-order and post-order.

Week 10:

- Write a program for BFS traversal on given graph.
- Write a program for DFS traversal on given graph.

Week 11:

- Write a program to implement Strassen's Matrix Multiplication
- Write a program to implement Job Sequencing with deadlines
- Write a program to implement Fractional knapsack problem

Week 12:

- Write a program to implement 0/1 knapsack
- Write a program to implement Travelling Sales Person Problem.

Outcomes: Upon the successful completion of the course, the student will be able:

- Implement and analyze various searching and sorting algorithms such as linear search, binary

search, bubble sort, selection sort, insertion sort, and quicksort.

2. Implement data structures such as hashing, stacks, queues, linked lists, binary trees, and graphs.
3. Apply greedy method, dynamic programming, backtracking, and branch and bound techniques to solve various problems such as matrix multiplication, job sequencing, knapsack problems, shortest path, and graph problems

Textbooks:

1. Data Structures, Second Edition, Richard F Gilberg, Forouzan, Cengage.
2. Fundamentals of Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekharan, Galgotia publications Pvt. Ltd

References:

1. Data Structures and Algorithms Made Easy, Narasimha Karumanchi.
2. Data Structures using C, Aaron M. Tanenbaum, Yedidiah Langsam, Moshe J Augenstein, Pearson.
3. Data structures and Algorithm Analysis in C, Mark Allen Weiss, Pearson Education. Ltd., Second Edition.
4. Design and Analysis of Algorithms, Aho, Ullman and Hopcroft, Pearson Education.

DATABASE MANAGEMENT SYSTEMS LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A643E	PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites:

Objectives: The Students will learn the following:

- Foundation knowledge in database concepts, technology and practice to groom students into well informed database application developers.
- Strong practice in SQL programming through a variety of database problems.
- Develop database applications using front-end tools and back-end DBMS
- Interpret the knowledge on database design.
- Determine the knowledge on key constraints and Normalization.
- Determine the knowledge on triggers, procedures, cursors and functions

List of Experiments/Sheets:**Experiment 1: Working with ER Diagram and Normalization**

Example: ER Diagram for Sailors Database Entities:

1. Sailor
2. Boat

Relationship: Reserves

Primary Key Attributes:

1. SID (Sailor Entity)
2. BID (Boat Entity)

Note:

Analyze the given data carefully and come up with the entities in it. Identify what data has to be persisted in the database. This contains the entities, attributes etc. Identify the primary keys for all the entities. Identify the other keys like candidate keys, partial keys, if any.

The student is to submit a document by writing the entities, relationships and key to the lab faculty.

Experiment 2: Working with DDL, DML, DCL and Key Constraints Creation, Altering and Dropping of Tables and Inserting rows into a Table (Use Constraints While Creating Tables) Examples Using Select Command.

Experiment 3: Working with Queries and Nested QUERIES Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, Constraints

Experiment 4: Working with Queries USING Aggregate Operators & views Queries using Aggregate Functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and Dropping of Views

Experiment 5: Working with Conversion Functions & String Functions Queries using Conversion Functions (to_char, to_number and to_date), String Functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), Date Functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)

Experiment 6: Working with Triggers using PL/SQL Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers

Experiment 7: Working with PL/SQL Procedures Programs Development using Creation of Procedures, Passing Parameters IN and OUT of PROCEDURES

Experiment 8: Working with LOOPS using PL/SQL and Exception Handling Program Development using WHILE LOOPS, Numeric FOR LOOPS, Nested Loops using ERROR Handling, BUILT-IN Exceptions, USE Defined Exceptions, RAISE- APPLICATION ERROR

Experiment 9: Working with Functions Using PL/SQL Program Development using Creation of Stored Functions, Invoke Functions in SQL Statements and Write Complex Functions.

Experiment 10: Working CURSORS Develop Programs using Features Parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of Clause and CURSOR Variables

Experiment 11: Installation of MYSQL and verification of online platforms to support creation of database through online mode.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply DDL, DML, DCL Commands using SQL.
2. Construct an Entity Relationship model for any large Enterprise.
3. Apply key constraints to the relational model.
4. Apply triggers, procedures and functions using PL/SQL
5. Develop Cursors for dynamic database applications

Textbooks:

1. Database System Concepts, Abraham Silberschatz, Henry F. Korth, S. Sudharshan, 6th Edition, Tata McGraw Hill, 2011.
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, McGrawHill Education, 3rd Edition, 2003.

References:

1. An Introduction to Database Systems, C. J. Date, A.Kannan, S. Swamynathan, 8th Edition, Pearson Education, 2006.
2. Fundamentals of Database Systems, Ramez Elmasri, Shamkant B. Navathe, 6th Edition, Pearson, Education, 2011.
3. Database Management Systems, G. K. Gupta, Tata McGraw Hill, 2011.

ENVIRONMENTAL SCIENCE AND RURAL SENSITIZATION

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-I	A634N	MS	L	T	P	C	CIE	SEE	Total
				2	-	-	-	-	-	-

Pre-requisites: Computational Thinking and Programming, Python Programming

Objectives: The Students will learn the following:

- To learn the fundamentals of database systems and analyze the data models to represent a database system using ER diagrams.
- To learn the Relational Model using SQL and formulate basic SQL Queries.
- To learn the Schema Refinement approaches and functional dependencies for effective Database Design.
- To Learn the Normalization rules for effective database design.
- To learn the Importance of Transactions, Concurrency Control mechanism to real-time environment and analyze the importance of recovery management
- To learn the internal storage structures using different file and indexing techniques which will help in physical Database design.

Unit I: (08 Lectures)

Ecosystem and Biodiversity Introduction to Ecosystem, types and functions of ecosystem, Value of biodiversity, Conservation of biodiversity: Insitu conservation & Ex. situ conservation

Activity:

- Studying the nearest lake water/farm land/forest Eco systems and preparing document and creating awareness programs among public in mitigating pollution.
- Poster presentation on nutrient cycles.

Unit II: (10 Lectures)

Energy & Water Resources

Introduction: Non-conventional energy resources (solar/wind/tidal) Water management (rain water harvesting, watershed management and water conservation) Global Environmental Challenges: Global warming and climate change, acid rains, ozone layer depletion. Solid Waste Management: Sources, classification, effects and control measures of urban and industrial solid wastes

Activity:

- Conducting mass level cycling for making aware of public in minimizing use of fossil fuels (ECO CLUBACTIVITY).
- To prepare rain water harvesting models.
- To demonstrate the generation of electricity with the utilization of non-conventional energy resource.

Unit III: (10 Lectures)

Green Chemistry & Hazardous Chemicals: Green Chemistry: Introduction & Principles. Classification of hazardous chemicals, transportation of hazardous chemicals, Hazchem code, Storage and handling of hazardous substances, Emergency preparedness (on site & offsite), Safety audit, Concept of fire and explosion, Major accidents involving hazardous substances, e-waste management

Activity:

- To make the flow chart on any one project
- Mass level plantation programs and involving students in Government programs.
- e-waste management.

Unit IV: (10 Lectures)

Empowering Villages: Jugaad –adapting latest developments- rural technologies –strategies-value addition-marketing the products-types of soils, important features of agriculture- vegetable crops-field crops-horticulture-fruit crops-flowery culture- green house-poly house.

Activity: Field visits and Interaction with farmers.

Unit V: (10 Lectures)

Health and Hygiene: Vitamins (definition& classification)-deficiency disease –balanced and nutrition diet-Awareness of contagious diseases-Epidemics-endemics-pandemics

Activity: Visit to local health centers.

Outcomes: Upon the successful completion of the course, the student will be able:

- Interpret the importance of functions of ecosystem & conservation of biodiversity.
- Choose the correct methods to reduce Global problems.
- Identify the hazardous chemicals.

4. Outline role of technology in empowering villages.
5. Create better awareness on health and hygiene.

Textbooks:

1. A Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for UGC 2014.
2. Environmental Studies by R. Raja gopalan, 2nd Edition, 2011, Oxford University Press.

References:

1. A Textbook of Environmental Studies by Shaashi Chawla, TMH, New Delhi 2011.
2. A Textbook of Environmental Studies by Anubha Kaushik, Fourth Edition, New Age International Publishers, 2014
3. Text book of Environmental Science & Technology – Dr.M. Anji Reddy, 2007, BS Publications.
4. Text Book of Environmental Studies by Deeshita Dave & P. Udaya Bhaskar, Cengage Learning 2008
5. Environmental Studies by K.V.S.G. Murali Krishna, VGS Publishers, Vijayawada. 2012
6. Environmental Studies by Benny Joseph, Tata McGraw Hill Co, New Delhi. 2008

References:

1. An Introduction to Database Systems, C. J. Date, A.Kannan, S. Swamynathan, 8th Edition, Pearson Education, 2006.
2. Fundamentals of Database Systems, Ramez Elmasri, Shamkant B. Navathe, 6th Edition, Pearson, Education, 2011.
3. Database Management Systems, G. K. Gupta, Tata McGraw Hill, 2011

II YEAR II SEM OPERATING SYSTEMS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A64KA	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites:**Objectives:** The Students will learn the following:

1. Analyze the tradeoffs inherent in operating system design.
2. Summarize various approaches to solve the problem of process concurrency in an operating system.
3. Evaluate the memory usage trade-offs in terms of size (main memory, processor speed).
4. Understand disk storage strategies and file strategies with protection and security issues.

Unit I:**(09 Lectures)****Introduction:** Concept of Operating Systems (OS), Generations of OS, Types of OS, OS Services, Interrupt handling and System Calls, Basic architectural concepts of an OS, Concept of Virtual Machine, Resource Manager view, process view and hierarchical view of an OS.**Processes:** Definition, Process Relationship, Different states of a Process, Process State transitions, Process Control Block (PCB), Context switching.**Thread:** Definition, Various states, Benefits of threads, Types of threads, Concept of multithread.**Unit II:****(10 Lectures)****Process Scheduling:** Foundation and Scheduling objectives, Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time.**Scheduling algorithms:** Pre-emptive and non-pre-emptive, FCFS, SJF, RR; Multiprocessor scheduling: Real Time scheduling: RM and EDF.**Inter-process Communication:** Concurrent processes, precedence graphs, Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Semaphores, Strict Alternation, Peterson's Solution, The Producer / Consumer Problem, Event Counters, Monitors, Message Passing, Classical IPC Problems: Reader's & Writer Problem, Dining Philosopher Problem, Barber's shop problem.**Unit III:****(10 Lectures)****Deadlocks:** Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.**Concurrent Programming:** Critical region, conditional critical region, monitors, concurrent Languages, communicating sequential process (CSP); Deadlocks - prevention, avoidance, detection and recovery.**Memory Management:** Basic concept, Logical and Physical address maps, Memory allocation: Contiguous Memory allocation - Fixed and variable partition-Internal and External fragmentation and Compaction.**Unit IV:****(10 Lectures)****I/O Hardware:** I/O devices, Device controllers, Direct Memory Access, Principles of I/O.**File Management:** Concept of File, Access methods, File types, File operation, Directory structure, File System structure, Allocation methods (contiguous, linked, indexed), Free-space management (bit vector, linked list, grouping), directory implementation (linear list, hash table), efficiency and performance.**Unit V:****(09 Lectures)****Disk Management:** Disk structure, Disk scheduling - FCFS, SSTF, SCAN, C-SCAN, Disk reliability, Disk formatting, Boot block, Bad blocks.**Case study:** UNIX OS file system, shell, filters, shell programming, programming with the standard I/O, UNIX system calls.**Outcomes:** Upon the successful completion of the course, the student will be able:

1. Understand the basic concepts of OS, processes and threads.
2. Identify and evaluate process scheduling criteria of OS.
3. Develop procedures for process synchronization and concurrent programming for avoiding deadlocks.
4. Distinguish disk access, file systems supported by an OS.
5. Extend operating systems disk scheduling techniques and system calls .

Textbook:

1. Operating System Concepts Essentials, Abraham Silberschatz, Peter Baer Galvin and Greg Gagne.

References:

1. Operating Systems: Internals and Design Principles. William Stallings.
2. Operating System: A Design-oriented Approach. Charles Patrick Crowley.
3. Operating Systems: A Modern Perspective. Gary J. Nutt.
4. Design of the Unix Operating Systems. Maurice J. Bach.
5. Understanding the Linux Kernel, Daniel Pierre Bovet, Marca Cesati.

OBJECT ORIENTED PROGRAMMING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A64L1	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Computational Thinking and Programming

Objectives: The Students will learn the following:

1. To understand the basic concepts and fundamentals of platform independent object oriented language.
2. To Compare Tightly Coupled and Loosely Coupled Programming.
3. To demonstrate skills in writing programs using exception handling techniques and multithreading.
4. To Utilize the built-in Data Structures and different JDBC Drivers.
5. To use GUI components from AWT and Swings packages.

Unit I:**(11 Lectures)**

OOP concepts - Data abstraction, encapsulation, inheritance, benefits of inheritance, polymorphism, classes and objects, procedural and Object oriented programming paradigms.

Java Programming -History of Java, comments, datatypes, variables, constants, scope and life time of variables, operators, expressions, type conversion and casting, enumerated types, control flow, conditional statements, loops break and continue statements. simple java program, arrays, console input and output, formatting output, constructors, methods, static fields and methods, access control, this keyword, overloading methods and constructors, garbage collection, building strings, exploring string class.

Unit II: Inheritance and Packages**(11 Lectures)**

Inheritance: Inheritance Hierarchies Super and Sub Classes, Member Access Rules, Super Keyword, And Preventing Inheritance: Final Classes and Methods.

Polymorphism: Dynamic Binding, Method Overloading, Abstract Classes and Methods.

Interface: Interface vs. Abstract Classes, defining an Interface, Implementing Interfaces, Accessing Implementations Through Interfaces References, Extending Interface.

Packages: Defining, Creating and Accessing a Package, Understanding Class path, Importing Packages.

Unit III: Exception Handling and Multithreading**(11 Lectures)**

Exception Handling: Dealing with Errors, Benefits of Exception Handling, The Classification of Exceptions, Exception Hierarchy, Checked Exceptions and Unchecked Exception, Usage of Try, Catch, Throw, Throws, and Finally, Re-Throwing Exceptions, Creating Own Exception Subclasses.

Multithreading: Difference Between Multiple Processes and Multiple Threads, Thread States, Creating Threads, Interrupting Threads, Thread Priorities, Synchronizing Threads, Inter-Thread Communication.

Unit IV: Collections and JDBC**(12 Lectures)**

Collection Framework in Java: Introduction to Java Collections, Overview of Java Collection Framework, Generics, commonly used Collection Classes-Array List, Vector, Hash Table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, Calendar and Properties.

Files: Streams-Byte Streams, Character Streams, Text Input/output, Binary Input/output, Random-access File Operations, File Management using File Class.

Unit V: GUI using Java**(12 Lectures)**

Event Handling: Events, Event Sources, Event Classes, Event Listeners, Relationship Between Event Sources and Listeners, Delegation Event Model, Examples: Handling a Button Click, Handling Mouse Events, Adapter Classes.

GUI Programming with Java: The AWT Class Hierarchy, Introduction to Swing, Swing vs. AWT, Hierarchy for Swing Components, Containers-JFrame, JApplet, JDialog, JPanel, Overview of Some Swing Components, JButton, JLabel, JTextfield, JTextarea, Simple Swing Applications, Layout Management-Layout Manager Types- Border Grid and Flow.

Connecting to Database: JDBC Type I to IV Drivers, connecting to a Database, querying a Database and Processing the Results, Updating Data with JDBC.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Use the syntax and semantics of java programming language and basic concepts of OOP.
2. Develop reusable programs using the concepts of inheritance, polymorphism, interfaces and packages.
3. Apply the concepts of Multithreading and Exception handling to develop efficient and error free codes.
4. Utilize the built-in Data Structures and JDBC Drivers to handle large data sets.

5. Design event driven GUI and web related applications which mimic the real world scenarios.

Textbook:

1. Java Fundamentals: A Comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH.

References:

1. Java for Programming, P.J.Dietel Pearson Education.
2. Object Oriented Programming through Java, P.Radha Krishna, Universities Press.
3. Thinking in Java, Bruce Eckel, Pearson Education
4. Programming in Java, S.Malhotra and S.Choudhary, Oxford University Press.

MACHINE LEARNING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A64LA	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics

Objectives: The Students will learn the following:

1. Understand the fundamentals of Machine Learning and its various learning paradigms.
2. Comprehend the Statistical Decision Theory and its application in Regression and Classification models.
3. Analyze the Bias-Variance trade-off and its significance in Machine Learning models.
4. Build and interpret Regression and Classification models such as Linear Regression, Logistic Regression, and Naïve-Bayes.
5. Understand and implement different clustering algorithms such as Partitional Clustering, Hierarchical Clustering, and Density-based Clustering.

Unit I:**(09 Lectures)**

Introduction to Machine Learning: Brief Introduction, Learning Paradigms, Statistical Decision Theory – Regression and Classification, Bias-Variance.

Unit II:**(10 Lectures)**

Regression and Classification Models: Linear Regression and Multivariate Regression, Linear Classification, Logistic Regression, Principal Component Analysis and Linear Discriminant Analysis, Multiclass Classification, Naïve-Bayes, Bayesian Networks, Support Vector Machines.

Unit III:**(10 Lectures)**

Decision Trees: Regression Trees, Stopping criteria and Pruning, Categorical Attributes, ID3 and CART.

Clustering: Partitional Clustering, Hierarchical Clustering, Density based Clustering, The BIRCH Algorithm, The CURE Algorithm.

Unit IV:**(10 Lectures)**

Bootstrapping and Cross Validation: Class Evaluation Measures, ROC Curve, Validation Splits.

Ensemble Methods: Bagging, Stacking, Boosting, Gradient Boosting, Random Forest.

Unit V:**(09 Lectures)**

Neural Networks: Introduction, Perceptron Learning, Backpropagation, Training and Validation, Parameter Estimation – MLE, MAP, Bayesian.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply the basic principles and concepts of Machine Learning and its various learning paradigms.
2. Apply Statistical Decision Theory in Regression and Classification models to make informed decisions.
3. Implement Bias-Variance trade-off and its importance in Machine Learning models.
4. Interpret Regression and Classification models such as Linear Regression, Logistic Regression, and Naïve-Bayes.
5. Implement clustering algorithms such as Partitional Clustering, Hierarchical Clustering, and Density-based Clustering, and develop an understanding of their applicability in real-world scenarios.

Textbook:

1. Introduction to Machine Learning, Ethem Alpaydin 2nd Edition, MIT Press 2000
2. Machine Learning, Tom M. Mitchell, McGraw hill, 1997, ISBN: 0-07-042807-7.

References:

1. Hands-on Machine Learning with Scikit Learn, Keras, and Tensorflow: Concepts, tools, and techniques to build intelligent systems, Geron, Aurelien, O'Reilly Media, 2019.
2. Machine Learning using Python, Pradhan, Manaranjan, U. Dinesh Kumar, Wiley, IIM Bangalore, 2019.

AUTOMATA AND COMPILER DESIGN

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A64L2	PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites:

Objectives: The Students will learn the following:

1. Identify the definitions of Formal language of finite automata techniques.
2. Interpret concepts of regular sets and decision algorithms for context free grammars.
3. Design of the Push Down Automata and its equivalence.
4. To understand and list the different stages in the process of compilation.
5. Design top-down and bottom-up parsers
6. Develop algorithms to generate code for a target machine.

Unit I:**(09 Lectures)**

Finite Automata and Regular Expressions: Alphabet, Strings Language, Operations, Finite State Machine, Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA) -Formal definition, simpler notations (state transition diagram, transition table), language of a DFA and NFA. Equivalence of Deterministic and Nondeterministic Finite Automata, NFA to DFA Conversion, Applications of Finite Automata.

Regular Expressions and Regular Grammars: Regular sets, Regular expressions, identity rules, manipulation of regular expressions, equivalence between RE and FA, Regular Grammars – Equivalence between Regular Grammar and FA.

Unit II:**(10 Lectures)**

Context Free Grammars: Context Free Grammars, Derivations – LMD, RMD, Derivation Trees, Normal Forms – Chomsky Normal Form and Griebach Normal Form.

Pushdown Automata: Definition, Language of PDA, Equivalence of PDA's and CFG's, Deterministic Pushdown Automata.

Unit III:**(10 Lectures)**

Compiler Design: Language Processors, Phases of Compiler, The Role of Lexical Analyzer, Input Buffering, Specification and Recognition of Tokens.

Top Down Parsing: Recursive-Descent Parsing, Non- recursive Predictive Parsing; First and Follow, LL(1) Grammars.

Unit IV:**(10 Lectures)**

Bottom Up Parsing: Reductions, handle pruning, LR- SLR, Powerful LR Parsers – CLR and LALR Parsers, Operator Precedence Parser.

Intermediate Code Generation: Syntax Trees, Three-Address Code, Type Checking, Control Flow: Boolean Expressions, Flow-of-Control Statements.

Unit V:**(09 Lectures)**

Code Optimization: The Principal Sources of Optimization, Data Flow Analysis, Constant Propagation, Loops in Flow Graphs.

Code Generation: Issues in the Design of a Code Generator, Object Code Forms, Basic Blocks and Flow Graphs, Machine Dependent Optimization – Peephole Optimization, Register Allocation and Assignment.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Design an abstract machine (Finite Automata, Push Down Automata)
2. Determine a grammar for given languages.
3. Adapt compiler construction tools and the functionality of each stage of compilation process.
4. Apply different methods of lexical analysis in design of top-down parser
5. Determine a given parser for bottom-up parsing and describe the techniques for run time storage, code optimization and code generation.

Textbook:

1. Introduction to Automata Theory, Languages and Computation, Hopcroft, Motwani, and Ullman, Pearson Publishers, Third Edition, 2006.
2. Compilers Principles, Techniques and Tools Aho, Ullman, Ravisethy, Pearson Education.

References:

1. Introduction to Formal Languages and Automata Theory and Computation-Kamala Krithivasan and Rama R, Pearson.
2. Modern Compiler Construction in C, Andrew W.Appel Cambridge University Press.
3. Elements of Compiler Design, A.Meduna, Auerbach Publications, Taylor and Francis Group.

OBJECT ORIENTED PROGRAMMING LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A641L	PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites: Computational Thinking and Programming

Objectives: The students will learn the following:

1. Infer the knowledge of Java compiler and eclipse platform
2. Gain hands on experience with Java programming
3. Gain knowledge on developing Java database applications
4. To understand building Java GUI based applications using swing.

List of Programs:

Week-1:

1. Write a program to display default values of all primitive data types.
2. Write a program to give the example for 'this' operator. And also use the 'this' keyword as a return statement.
3. Write a program to demonstrate static variables, methods, and blocks.

Week-2:

1. Write a program to give the example for 'super' keyword.
2. Write a program that illustrates simple inheritance.
3. Write a program that illustrates multi-level inheritance.
4. Write a program demonstrating the method overloading and method overriding.
5. Write a program demonstrating constructor overloading.

Week-3:

1. Write a program that describes exception handling mechanisms.
2. Write a program to illustrate sub class exception precedence over base class.
3. Write a program for creation of user defined exceptions.

Week-4:

1. Write a program to illustrate creation of threads using runnable class.
2. Write a program that implements a multi-thread application that has three threads. First thread generates a random integer every 1 second and if the value is even, the second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number.

Week-5:

1. Write a program illustrating multiple inheritance using interfaces.
2. Write a program for Abstract Class and Interfaces.

Week-6:

1. Write a program to create a package named pl, and implement this package in ex1 class.
2. Write a Program for Implementation of Access Modifiers using various levels of Packages.
3. Write a Program Using String Tokenizer.

Week-7:

1. Write a program that displays a number of characters, lines and words in a text file.
2. Write a program based on File Handling Methods (Reading, Copying, Merging files).

Week-8:

1. Write a program that connects to a database using JDBC and does add, delete, modify and retrieve operations.

Week-9:

1. Program for handling the Mouse, Key and Window Events Implementation.

2. Write a Simple Program based on Collections (ArrayList, Sets, Trees, Queue, Stack and Vectors etc).

Week-10:

1. Write a program to Implement the various components in AWT (text area, textfield, buttons, Layout Control etc)

Week-11:

1. Write a program to Implement the various components in Swings (JFrame, JLabel, JAppletetc).

Week-12:

1. Write a program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -, *, /, % Operations. Add a text field to display the results. Handle any possible exceptions like divided by zero.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Develop the standalone applications for Multi-Threaded and Exceptions Handling.
2. Apply OOP in java programming in problem solving.
3. Ability to access data from a database with java programming.
4. Develop Java applets and applications
5. Develop Java applications using swing.

Textbooks:

1. Java Fundamentals: A Comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH.

References:

1. Java for Programming, P.J.Dietel Pearson Education.
2. Object Oriented Programming through Java, P.Radha Krishna, Universities Press.
3. Thinking in Java, Bruce Eckel, Pearson Education
4. Programming in Java, S.Malhotra and S.Choudhary, Oxford University Press.

MACHINE LEARNING LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A642L	PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The students will learn the following:

5. Make use of Data sets in implementing the machine learning algorithms
6. Implement the machine learning concepts and algorithms in Python language.

Week 1: Reading of different types of data files and preparing datasets into train and test datasets.

Week 2: Implement a program for **Linear regression**.

Week 3: Implement a program on **Logistic regression**.

Week 4: Write a program to implement the **Naïve Bayesian classifier**.

Week 5: Write a program to implement to cross validation using k-NN classifiers. Measure accuracy, recall, precision.

Week 6: Write a program to implement to cross validation using k-NN classifiers. Measure accuracy, recall, precision. Draw RoC.

Week 7: Write a program to construct a **SVM classifier**.

Week 8: Write a program to demonstrate the working of the **decision tree**.

Week 9: Write a program to implement **Random forest** for ensemble learning.

Week 10: Build an **Artificial Neural Network** by implementing the **Backpropagation algorithm**.

Week 11: Write a program to perform dimensionality reduction using **PCA**.

Week 12: Write a program to construct a **k-means clustering**.

Outcomes: Upon the successful completion of the course, the student will be able:

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin 2nd Edition, MIT Press 2000.
2. Machine Learning, Tom M. Mitchell, McGraw hill, 1997, ISBN: 0-07-042807-7.

References:

1. Hands-on Machine Learning with Scikit Learn, Keras, and Tensorflow: Concepts, tools, and techniques tom build intelligent systems, Geron, Aurelien, O'Reilly Media, 2019.
2. Machine Learning using Python, Pradhan, Manaranjan, U. Dinesh Kumar, Wiley, IIM Bangalore, 2019.

BUSINESS COMMUNICATION LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A641N	HS	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Course Objectives:

1. To inculcate the habit of reading short and critical texts.
2. To comprehend the content while listening.
3. To articulate thoughts and ideas eloquently.
4. To enable students to write grammatical and meaningful sentences.
5. To impart confidence and competence in expressing their views.

Experiment-I**Week -1 & 2:**

Listening: Listening to short conversations.

Speaking: Giving information about oneself and giving short talk on business related topics.

To gather information from conversations thereby deliver short talks based on business topics.

Week-3:

Reading: Reading short texts to understand the central idea/theme.

Writing: Writing a piece of internal business communication of 30-40 words (Email).

To read for central idea/theme in passages, write responses in business communication.

Experiment-II**Week-4 & 5:**

Listening: Listening to peer group and taking notes.

Speaking: Talking critically on business related topics and issues.

To inculcate and improve on note taking while listening, respond appropriately to critical business talks.

Week-6:

Reading: Reading graphs and business news by using sub-skills of reading.

Writing: Writing business proposals in 50-60 words.

To decode information from pictorial representation and get skilled in writing business proposals.

Experiment-III**Week-7:**

Listening: Listening to interviews and taking notes.

Speaking: Debates.

To observe the skills and answering techniques from Interview videos, participate in debates and presentations spontaneously.

Week-8:

Reading: Reading a longer text to understand and locate specific information.

Writing: Write a Business Letter or e-Mail of 60-80 words, based on an input text and some notes.

To overcome ambiguity while reading passages and write responses to business letters.

Experiment-IV**Week-9:**

Listening: Listening to songs and writing down the lyrics.

Speaking: Business Presentations.

To listen and write lyrics of English songs; and to make business presentations.

Week-10:

Reading: Read a longer text and answer questions.

Writing: Write a short play.

To comprehend longer passages and practice the creative skill of story writing.

Experiment-V**Week-11:**

Listening: Watching documentaries and making notes.

Speaking: Open Mike.

To practice the skills of making notes from documentaries and take opportunity confidently in public speaking.

Week-12:

Reading: Critical Reading to know author's perspective.

Writing: Writing a business report.

To read passages and understand the writer's intention and prepare business report.

Infrastructure:

- Classroom with a projector

Course Outcomes: The students will be able to:

1. Write business emails by comprehending the central idea and express opinions and respond in business scenarios.
2. Make speeches on business related topics, analyze and respond to the graphical texts in written and spoken contexts.
3. Perform well in debates and take notes effectively while listening.
4. Make business presentations effectively and write short stories.
5. Critically analyze the text and report effectively in business scenarios:

Suggested Reading:

1. Cambridge University Press India Pvt. Ltd, New Delhi.
2. Mandal S.K(2006), Effective Communication &Public Speaking, Jaico Publishing House, New Delhi.
3. Grant Taylor (2004), English Conversation Practice, Tata McGraw Hill, New Delhi.
4. Balasubramanian.T(2000),A textbook of English Phonetics for Indian Student, MacMillan Publishers, India.
5. KamaleshSadanand,SusheelaPunitha(2008),SpokenEnglish:AfoundationCourse:Parts1& 2, New Delhi, Orient Longman Pvt. Ltd.

Web References:

1. www.cambridgeenglish.org.
2. www.esl-lab.com.

GENDER SENSITIZATION

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	II-II	A643N	MC	L	T	P	C	CIE	SEE	Total
				2	-	-	-	-	-	-

Course Description

This course offers an introduction to Gender Studies, an interdisciplinary field that asks critical questions about the meanings of sex and gender in society. The primary goal of this course is to familiarize students with key issues, questions and debates in Gender Studies, both historical and contemporary. It draws on multiple disciplines – such as literature, history, economics, psychology, sociology, philosophy, political science, anthropology and media studies – to examine cultural assumptions about sex, gender, and sexuality.

This course integrates analysis of current events through student presentations, aiming to increase awareness of contemporary and historical experiences of women, and of the multiple ways that sex and gender interact with race, class, caste, nationality and other social identities. This course also seeks to build an understanding and initiate and strengthen programmes combating gender-based violence and discrimination. The course also features several exercises and reflective activities designed to examine the concepts of gender, gender-based violence, sexuality, and rights. It will further explore the impact of gender-based violence on education, health and development.

Objectives of the Course

- To develop students' sensibility with regard to issues of gender in contemporary India.
- To provide a critical perspective on the socialization of men and women.
- To introduce students to information about some key biological aspects of genders.
- To expose the students to debates on the politics and economics of work.
- To help students reflect critically on gender violence.
- To expose students to more egalitarian interactions between men and women.

Learning Outcomes

- Students will have developed a better understanding of important issues related to gender in contemporary India.
- Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender. This will be achieved through discussion of materials derived from research, facts, everyday life, literature and film.
- Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.
- Students will acquire insight into the gendered division of labor and its relation to politics and economics.
- Men and women students and professionals will be better equipped to work and live together as equals.
- Students will develop a sense of appreciation of women in all walks of life.
- Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence.

Unit-I: Understanding Gender

Introduction: Definition of Gender-Basic Gender Concepts and Terminology-Exploring Attitudes towards Gender-Construction of Gender-Socialization: Making Women, Making Men- Preparing for Womanhood. Growing up Male. First lessons in Caste.

Unit – II: Gender Roles and Relations

Two or Many? -Struggles with Discrimination-Gender Roles and Relations-Types of Gender Roles Gender Roles and Relationships Matrix-Missing Women-Sex Selection and Its Consequences Declining Sex Ratio. Demographic Consequences-Gender Spectrum: Beyond the Binary

Unit – III: Gender and Labour

Division and Valuation of Labour-Housework: The Invisible Labor- “My Mother doesn’t Work.” “Share the Load.”-Work: Its Politics and Economics -Fact and Fiction. Unrecognized and Unaccounted work. -Gender Development Issues-Gender, Governance and Sustainable Development-Gender and Human Rights-Gender and Mainstreaming

Unit – IV: Gender-Based Violence

The Concept of Violence- Types of Gender-based Violence-Gender-based Violence from a Human Rights Perspective-Sexual Harassment: Say No!-Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading: “Chupulu”. Domestic Violence: Speaking Out! Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Thinking about Sexual Violence Blaming the Victim-“I Fought for my Life....”

Unit – V: Gender and Culture

Gender and Film-Gender and Electronic Media-Gender and Advertisement-Gender and Popular Literature-Gender Development Issues-Gender Issues Gender Sensitive Language-Gender and Popular Literature - Just Relationships: Being Together as Equals Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers. Rosa ParksThe Brave Heart.

Note: Since it is Interdisciplinary Course, Resource Persons can be drawn from the fields of English Literature or Sociology or Political Science or any other qualified faculty who has expertise in this field from engineering departments.

- **Classes will consist of a combination of activities: dialogue-based lectures, discussions, collaborative learning activities, group work and in-class assignments.**
- **Apart from the above prescribed book, Teachers can make use of any authentic materials related to the topics given in the syllabus on “Gender”.**

Essential Reading:

The Textbook, “Towards a World of Equals: A Bilingual Textbook on Gender” written by A.Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu published by Telugu Akademi, Telangana Government in 2015.

III YEAR I SEM DEEP LEARNING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Machine Learning

Objectives: The Students will learn the following:

1. Understand the fundamentals of deep learning, including historical trends, the perceptron, deep forward neural networks, and capacity, as well as concepts such as overfitting, underfitting, and regularization.
2. Understand convolutional networks, including the convolution operation, layers, filters, and pooling strategies. Students will also learn about different variants of convolution functions and how they can be used for tasks such as detection and segmentation.
3. Understand CNN architectures, including VGGNet, LeNet, AlexNet, ResNet, InceptionNet, MobileNet, and DenseNet.
4. Understanding sequence modeling, including recurrent neural networks, bidirectional RNNs, encoder-decoder architectures, deep recurrent networks, and LSTM and other gated RNNs. Students will also learn about optimization techniques.
5. Understand deep generative models such as deep belief networks, restricted Boltzmann machines (RBMs), generative adversarial networks (GANs), and autoencoders.

Unit I:

(09 Lectures)

Introduction to Deep Learning: History of Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Perceptrons, Perceptron Learning Algorithm, Multilayer Perceptrons (MLPs), Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, Feedforward Neural Networks, Representation Power of Feedforward Neural Networks, Backpropagation.

Unit II:

(10 Lectures)

Learning Rate: Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp, Adam, Eigenvalues and eigenvectors, Eigenvalue Decomposition.

Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout.

Unit III:

(10 Lectures)

CNN Architectures: Convolutions, Convolutional Layers, Pooling Strategies, LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Visualizing Convolutional Neural Networks, Guided Backpropagation, Deep Dream, Deep Art, Fooling Convolutional Neural Networks.

Unit IV:

(10 Lectures)

Sequence Learning: Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs. Encoder Decoder Models, Attention Mechanism, Attention over Images.

Unit V:

(09 Lectures)

Deep Generative Models: Deep Belief Networks, Restricted Boltzmann Machines (RBMs), Generative Adversarial Networks (GAN's), Auto encoders.

Transfer Learning: Approaches in Transfer Learning, Transfer Learning with Inception Model.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Identify and address issues such as overfitting and underfitting, and understand how regularization can be used in deep learning.
2. Apply convolutional networks to perform the convolution operation, understand layers and filters, apply parameter sharing and pooling strategies, and use different variants of the convolution function for tasks such as detection and segmentation.
3. Evaluate and use different CNN architectures, including VGGNet, LeNet, AlexNet, ResNet, InceptionNet, MobileNet, and DenseNet.
4. Apply sequence modeling techniques, including recurrent neural networks, bidirectional RNNs, encoder-decoder architectures, deep recurrent networks, and LSTM and other gated RNNs. Students will also learn about optimization techniques for these models.
5. Apply deep generative models such as deep belief networks, restricted Boltzmann machines (RBMs), generative adversarial networks (GANs), and autoencoders.

Textbooks:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, MIT Press.
2. Deep Learning from Scratch: Building with Python from First Principles, Seth Weidman, O'Reilly.

References:

1. Deep Learning for Beginners: A beginner's guide to getting up and running with deep learning from scratch using Python, Pablo Rivas, Packt Publishing Ltd, 2020.
2. Deep Learning with Tensorflow, Giancarlo Zaccane, Md. Rezaul Karim, Ahmed Menshawy.

COMPUTER NETWORKS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites:**Course Objectives:**

1. To illustrate the importance of layering and the functionality of various layers
2. To analyze the data detection and correction methods.
3. To make use of the various routing algorithms in the Internetwork.
4. To analyze the Transport protocols in the data transmission.
5. To utilize the different applications of Application Layer.

Unit I:**(10 Lectures)****Introduction to Networks:** Network Hardware: Network hardware: Topologies, Types of networks.**Network software:** Protocol Hierarchy, Protocols: OSI Reference Model, TCP/IP Reference Model.**Physical Layer:** Transmission Media: Guided: twisted pairs, coaxial cable, fiber optics and Wireless LAN. (T1 Ch5, T2: Ch1, Ch2)**Unit II:****(12 Lectures)****Data Link Layer:** DLL Services, Framing, Error Control: Parity bit method, CRC, Hamming code, and Flow Control.**Simplex Protocol:** simplex stop and wait protocol for an error-free channel, simplex stop and wait protocol for noisy channel. Sliding Window Protocol: One-bit sliding window protocol, Protocol using Go-Back-N, Protocol using Selective Repeat. (T1 Page 259, T2, Ch3)**Unit III:****(12 Lectures)****Medium Access Control Sub Layer:** Multiple access protocols: ALOHA, CSMA, Collision free protocols, HDLC and Switching Techniques.**Network Layer:** Logical addressing – IPV4, IPV6, Address Mapping – ARP, RARP, DHCP, ICMP, IGMP.**Routing algorithms:** Shortest Path Routing, Flooding, Hierarchical routing, Broadcast, Multicast, Distance Vector Routing, Internetworking. (T1 Page 326, Page 596, T2: Ch4, Ch5)**Unit IV:****(10 Lectures)****Transport Layer:** Transport Services, Connection Management using three-way handshake principle, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), SCTP, Congestion Control Policies, QoS Techniques: Leaky Bucket and Token Bucket algorithm. (T1 Page 693, T2 Ch 6)**Unit V:****(08 Lectures)****Application Layer:** DNS, TELNET, E-MAIL, FTP, WWW, HTTP, SNMP, Streaming audio and video. (T1 Page 995, T2 Ch 7)**Course Outcomes:**

1. List out various Network Reference models and transmission media.
2. Illustrate Data Link Layer design issues and protocols.
3. Model the routing algorithms in various types of networks.
4. Make use of suitable protocols for ensuring QoS in data transportation.
5. Dissect and utilize the user-defined applications in real-time scenarios.

Text Books:

1. Data Communication and Networking, 4th Edition, Behrouz A. Forouzan, McGraw- Hill.

Reference Books:

1. Computer Networks, 8th Edition, Andrew S. Tanenbaum, Pearson New International Edition.
2. Data and Computer Communication, 8th Edition, William Stallings, Pearson Prentice Hall India.
3. Internetworking with TCP/IP, Volume 1, 6th Edition Douglas Comer, Prentice Hall of India.

DATA MINING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Probability and Statistics, Database Management Systems

Course Objectives:

1. To learn introductory and necessity of Data Mining and Data Warehousing.
2. To understand the importance of data pre-processing for data analysis.
3. To learn in depth about the association rules and to understand frequent item sets
4. Ability to classify the data into various formats and predicting the unknown data
5. To learn grouping of data objects without knowing class labels in prior.

Unit I: Data Warehousing and ETL**(10 Lectures)**

Introduction to Data Warehouse, OLTP vs. OLAP, Data Warehouse Architecture, Data Warehouse, ETL Process, ELT and Data Lakes, ETL Tools, Modelling: Multidimensional Data Model, DW Schemas, Data Warehouse Implementation, From Data Warehousing to Data Mining.

Unit II: Data Pre-processing**(08 Lectures)**

Needs of Pre-processing the Data, Data Cleaning, Data Integration and Transformation, Data Reduction, Discretization and Concept Hierarchy Generation.

Unit III: Data Mining**(08 Lectures)**

Fundamentals of data mining, Knowledge Discovery from Databases (KDD), Data Mining Functionalities, Classification of Data Mining systems, Data Mining Primitives, Major issues in Data Mining, Data Mining applications and tools.

Unit IV: Unsupervised Learning**(10 Lectures)**

Association Rule Mining: Basic Concepts, Frequent Item set Mining Methods, Mining Multilevel Association Rules, Mining Multidimensional Association Rules, From Association Mining to Correlation Analysis, Constraint Based Association Mining.

Cluster Analysis: Introduction, Types of Data in Cluster Analysis, A Categorization of Major Clustering Methods- Partitioning Methods, Hierarchical Methods, Density-Based Methods, Outlier Analysis.

Unit V: Supervised Learning**(12 Lectures)**

Classification and Prediction: Issues Regarding Classification and Prediction, Classification by Decision Tree Induction, Bayesian Classification, Classification by Back propagation, Other Classification Methods, Evaluating the Accuracy, Techniques to Improve Classification Accuracy

Regression Analysis: Definition, regression types, regression methods, Linear Regression Models and Least Squares, Shrinkage Methods, Case Studies.

Course Outcomes:

1. Illustrate multi- dimensional data model by understanding the characteristics of a data warehouse.
2. Infer data pre-processing steps to prepare the data ready for analysis.
3. Choose appropriate data mining techniques to mine the interesting patterns from the data.
4. Apply unsupervised learning algorithms to find frequent patterns and clusters on real world data.
5. Compare various classification techniques and analyse their accuracy in determining class labels.

Text Books:

1. Data Mining – Concepts and Techniques - Jiawei Han & Micheline Kamber, Morgan Kaufmann Publishers.
2. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Pearson.

Reference Books:

1. Data Mining Introductory and advanced topics –Margaret H Dunham, Pearson Education.
2. Data Mining Techniques – Arun K Pujari, University Press.
3. Data Warehousing Fundamentals – Paulraj Ponnaiah Wiley Student Edition.

FUNDAMENTALS OF DESIGN THINKING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Probability and Statistics, Database Management Systems

Course Objectives:

1. To learn introductory and necessity of Data Mining and Data Warehousing.
2. To understand the importance of data pre-processing for data analysis.
3. To learn in depth about the association rules and to understand frequent item sets
4. Ability to classify the data into various formats and predicting the unknown data
5. To learn grouping of data objects without knowing class labels in prior.

Unit I:**(08 Lectures)**

Introduction: Design Thinking-Definition, Need of Design Thinking, Variety within the Design Thinking Discipline, phases of Design Thinking, Design Thinking Mindset Tools for Design Thinking, Relevance of Design and Design Thinking in Engineering.

Unit II:**(08 Lectures)**

Approach for Design Thinking: Fundamental Concepts - Empathy, Ethnography, Divergent Thinking, Convergent Thinking, Visual Thinking, Assumption Thinking, Prototyping, learning and validation.

Unit III: Data Mining**(08 Lectures)**

Thinking Resources and Processes: Resources - People, Place, Materials, Organizational Fit. Processes - Numerous Approaches, Double Diamond Process, 5 Stage d.school Process, Designing for growth process

Idea Generation: Basic design direction-Themes of thinking-Brainstorming- value addition-sketching-presenting ideas.

Unit IV: Unsupervised Learning**(08 Lectures)**

Implementation of Design Thinking: Format selection – Material selection – finishing – techniques for finishing-classification; advanced technology for marketing – scale-continuity.

Unit V: Supervised Learning**(08 Lectures)**

Application of Design Thinking: Design Thinking in Various Sectors -Health Care, Finance, Education, Infrastructure, Information Technology, Manufacturing; Design thinking case studies in engineering and management.

Course Outcomes:

1. Understand the different phases of design thinking
2. Implement the design thinking using different approaches
3. Exemplify the importance of resources for design thinking
4. Evaluate the different themes of design thinking
5. Apply the techniques for design thinking
6. Practice design thinking for engineering and management.

Text Books:

1. Change by Design: How Thinking Transforms Organizations and Inspires Innovation, Tim Brown, 1st Edition, Harper Collins, 2009.
2. Design thinking for Strategic innovation, Idris Moote, Wiley, 2013.

Reference Books:

1. Design Thinking, Gavin Ambrose and Paul Harris, AVA Publishing, 2010.
2. Design Thinking, Christoph Meinel and Larry Leifer, Springer, 2011.
3. This is Service Design Thinking, Marc stickdorn and Jacob Schneider, Wiley, 2011.

**SOFT COMPUTING
(PROFESSIONAL ELECTIVE I)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-I	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Linear Algebra and Calculus.

Course Objectives:

1. To understand mathematical computational techniques, such as neural networks, genetic algorithms, and hybrid systems, to explore their applications in soft computing and artificial intelligence domains.
2. To understand various types of supervised learning networks and their architectures.
3. To understand and develop algorithms for pattern association in neural networks, enabling autoassociative and heteroassociative memory capabilities as well as bidirectional and temporal associations and unsupervised learning networks to learn feature extraction in neural networks.
4. To explore and understand various specialized neural network architectures to leverage their unique characteristics for solving specific problems in artificial intelligence and computational neuroscience.
5. To understand self-organized maps and their application into clustering and investigate stability conditions of a specific class of non-linear systems.

Unit I:

(09 Lectures)

Introduction: Neural Networks, Application Scope of Neural Networks, Fuzzy Logic, Genetic Algorithm, Hybrid Systems, Soft Computing

Artificial Neural Network: An Introduction, Fundamental Concept, Evolution of Neural Networks, Basic Models of Artificial Neural Network, Important Terminologies of ANNs, McCulloch–Pitts Neuron, Linear Separability, Hebb Network.

Unit II:

(09 Lectures)

Supervised Learning Network: Perceptron Networks, Adaptive Linear Neuron (Adaline), Multiple Adaptive Linear Neurons, Back-Propagation Network, Radial Basis Function Network, Time Delay Neural Network, Functional Link Networks, Tree Neural Networks, Wavelet Neural Networks.

Unit III:

(11 Lectures)

Associative Memory Networks: Training Algorithms for Pattern Association, Autoassociative Memory Network, Heteroassociative Memory Network, Bidirectional Associative Memory (BAM), Hopfield Networks, Iterative Autoassociative Memory Networks, Temporal Associative Memory Network.

Unsupervised Learning Networks: Fixed Weight Competitive Nets, Kohonen Self-Organizing Feature Maps, Learning Vector Quantization, Counterpropagation Networks, Adaptive Resonance Theory Network.

Unit IV:

(09 Lectures)

Special Networks: Simulated Annealing Network, Boltzmann Machine, Gaussian Machine, Cauchy Machine, Probabilistic Neural Net, Cascade Correlation Network, Cognitron Network, Neocognitron Network, Cellular Neural Network, Logicon Projection Network Model, Spatio-Temporal Connectionist Neural Network, Optical Neural Networks.

Unit V:

(10 Lectures)

Clustering of Self-Organizing Feature Maps: Concept of Clustering, Training of SOMs, Clustering of SOM: Method I, Clustering of SOM: Method II.

Stability Analysis of a Class of Artificial Neural Network Systems: Stability Conditions of a Class of Non-Linear Systems, Formation of Main Matrices and Sub-Matrices for an Artificial Neural Network System, Methodology Developed for Stability Analysis of Artificial Neural Networks.

Course Outcomes:

1. Comprehend mathematical computational techniques such as neural networks, genetic algorithms, and hybrid systems and apply them in soft computing and artificial intelligence domains.
2. Recognize different types of supervised learning networks and describe their architectures.
3. Construct algorithms for pattern association in neural networks, facilitating autoassociative and heteroassociative memory capabilities, as well as bidirectional and temporal associations, and to implement unsupervised learning networks for feature extraction in neural networks.
4. Categorize and appraise various specialized neural network architectures, and analyze their distinct characteristics to solve specific problems in artificial intelligence and computational neuroscience.

5. Explain the concept of self-organized maps, their application in clustering, and assess stability conditions of a specific class of non-linear systems.

Text Books:

1. Principles of Soft Computing, S N Sivanandam, S N Deepa, 3rd Edition, Wiley Publications.

Reference Books:

1. Artificial Neural Networks, B Yegnanarayana, PHI.
2. Neural Networks, Fuzzy Systems, and Evolutionary Algorithms: Synthesis and Applications, S. Rajasekaran, G.A. Vijayalakshmi Pai, PHI.

**GEOMETRY FOR COMPUTER VISION
(PROFESSIONAL ELECTIVE I)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-I	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Mathematics

Course Objectives:

1. To Provide a solid foundation in Projective Geometry and its applications in various transformations and estimations, with a focus on planar geometries and the 2D projective plane.
2. To Understand the principles and mathematical models underlying the process of capturing images with cameras and the geometric relationships between objects and their images in the captured scene.
3. To understand the mathematical principles and techniques involved in analyzing the relationship between two camera views.
4. To understand and analyze the relationships and properties of multiple views or perspectives of a scene or object.
5. To understand the auto-calibration techniques and the application of duality principles to address the challenges in calibrating cameras and reconstructing 3D scenes from images.

Unit I: (09 Lectures)

The Background: Projective Geometry, Transformations and Estimation: Planar geometry, The 2D projective plane, Projective transformations, A hierarchy of transformations, The projective geometry of 1D, Topology of the projective plane, Points and projective transformations, Representing and transforming planes, lines and quadrics, Twisted cubics, The hierarchy of transformations, The plane at infinity.

Unit II: (10 Lectures)

Camera Geometry and Single View Geometry : Finite cameras, The projective camera, Cameras at infinity, Other camera models, Closure, Basic equations, Geometric error, Restricted camera estimation, Radial distortion, Action of a projective camera on planes, lines, and conics, Images of smooth surfaces, Action of a projective camera on quadrics, The importance of the camera centre, Camera calibration and the image of the absolute conic, Vanishing points and vanishing lines, Affine 3D measurements and reconstruction.

Unit III: (10 Lectures)

Two-View Geometry : Epipolar geometry, The fundamental matrix F, Fundamental matrices arising from special motions, Geometric representation of the fundamental matrix, Retrieving the camera matrices, The essential matrix, Closure, Outline of reconstruction method, Reconstruction ambiguity, The projective reconstruction theorem, Basic equations, The normalized 8-point algorithm, The algebraic minimization algorithm, Geometric distance, Experimental evaluation of the algorithms, Automatic computation of F.

Unit IV: (08 Lectures)

N-View Geometry: Bilinear relations, Trilinear relations, Quadrilinear relations, Intersections of four planes, counting arguments, Number of independent equations, choosing equations, Closure, Projective reconstruction – bundle adjustment, Affine reconstruction – the factorization algorithm, Non-rigid factorization, Projective factorization.

Unit V: (08 Lectures)

Auto-Calibration and Duality: Introduction, Algebraic framework and problem statement, Calibration using the absolute dual quadric, The Kruppa equations, A stratified solution, Calibration from rotating cameras, Carlsson–Weinshall duality, Reduced reconstruction, Closure.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Illustrate the background concepts of Geometry using Transformations and Estimation.
2. Classify the Properties of Camera Geometry and Single View Geometry.
3. Determine about the Equations to Reconstruct Two View Geometry.
4. Apply Equations to reconstruct N-View Geometry.
5. Design Algebraic Solutions to Auto Calibration and Duality.

Textbooks:

1. Multiple View Geometry in Computer Vision, Richard Hartley and Andrew Zisserman, Second Edition, Cambridge.
2. The Four Pillars of Geometry, John Stillwell, Springer, 1st Edition.

References:

1. Perspective and Projective Geometry by Annalisa Crannell, Marc Frantz, and Fumiko Futamura.

**AUTOMATED REASONING AND THEOREM PROVING
(PROFESSIONAL ELECTIVE I)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-I	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Probability and Statistics

Course Objectives:

1. Be able to represent mathematical and other knowledge using the logical formalism
2. Understand the history of formalizing mathematical knowledge
3. Know and understand the advantages and limitations of the main approaches and techniques in automated reasoning of mathematical knowledge;
4. Be able to apply different automated reasoning techniques to new problems
5. Able to learn semantic Information to guide Proofs.

Unit I: (08 Lectures)

Introduction: The historical perspective: Mathematical Logic, Psychological studies, Automatic theorem proving.

Formal Notations: Arguments about Propositions, The Internal structure of propositions.

Unit II: (08 Lectures)

Arguments about Propositions: Truth functional connectives, Propositional Formula. The internal structure of Propositions: Functions of Predicates, variables and constants, the status of variables, the meaning of formulae.

Unit III: (08 Lectures)

Uniform Proof Procedures

Formalizing the Notion of Proof: The Resolution rule, Kowalski form, the paramodulation rule.

Criticisms of Uniform Proof Procedures: The contribution of Logic, A resolution proof and combinatorial explosion, attempt to guide search, analysing human proofs, a new methodology.

Unit IV: (08 Lectures)

Guiding Search Decision procedure for inequalities: Axioms for Inequalities, some human proofs, types, the sup-Inf method, variable elimination. Rewrite Rules: what is rewrite rules, some sample rewrite rule sets, termination, other important properties, applying rewrite rules.

Unit V: Supervised Learning (08 Lectures)

Using semantic Information to guide Proofs: formalizing geometry, geometric proofs, construction.

Applications of Artificial Mathematicians: Algebraic manipulation system, Automatic Theorem proving, computer-assisted instructions, Logic Programming.

Course Outcomes:

1. Able to understand mathematical and other knowledge using the logical formalism
2. Able to understand the history of formalizing mathematical knowledge
3. Students can know and understand the advantages and limitations of automated reasoning of mathematical knowledge
4. Be able to apply different automated reasoning techniques to new problems
5. Students able to learn semantic Information to guide Proofs.

Text Books:

1. Bundy, Alan. The computer modelling of mathematical reasoning. Vol. 10. London: Academic Press, 1983.

Reference Books:

1. Bundy, Alan. Artificial Mathematicians: The Computational Modelling of Mathematical Reasoning. The United Kingdom, Department of Artificial Intelligence, University of Edinburgh, 1981.
2. Huth, Michael, and Ryan, Mark. Logic in Computer Science: Modelling and Reasoning about Systems. N.p., Cambridge University Press, 2004.

Web References:

1. <https://www.cl.cam.ac.uk/teaching/1617/L18/>

**STATISTICAL ANALYSIS FOR DATA SCIENCE
(PROFESSIONAL ELECTIVE I)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-I	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics

Objectives: The Students will learn the following:

1. Understand the problem and various data collections techniques.
2. Describe data and its behavior using statistical measures.
3. Summarize the Data using data distribution techniques.
4. Understand random variable representation.
5. Identify how to distribute data and Input modeling.

Unit I: Statistics and the Scientific Method

(10 Lectures)

Introduction, Why Statistics. Some Current Applications of Statistics. Using Surveys and Experimental Studies to Gather Data: Introduction and Abstract of Research Study, Observational Studies, Sampling Designs for Surveys, Experimental Studies, Design for Experimental Studies.

Unit II: Data Description

(10 Lectures)

Describing data on a single variable: Graphical Methods, describing data on Single Variable: Measures of central tendency, describing data on single variable: Measures of variability, Summarizing Data from More than One variable; graphs and correlation.

Unit III: Random Variables and Probability Distributions

(10 Lectures)

Random Variables, Probability Distributions, Two or more Random variables, Conditional Distribution and Independence.

Unit IV: Random-Number Generation

(08 Lectures)

Properties of random numbers; Generation of pseudo-random numbers, Techniques for generating random numbers, Tests for Random Numbers, Random-Variate Generation: Inverse transform technique Acceptance-Rejection technique.

Unit V: Input Modelling

(08 Lectures)

Data Collection; Identifying the distribution with data, Parameter estimation, Goodness of Fit Tests, fitting a non-stationary Poisson process, Selecting input models without data, Multivariate and Time-Series input models.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Explain the nature of a given problem and describe different methods used for data collection.
2. Interpret and discuss the characteristics and behavior of data through various statistical measures.
3. Summarize and communicate the overall distribution of data using appropriate data distribution techniques.
4. Infer the concept of random variables and their representation in the context of data analysis.
5. Interpret methods of data distribution and input modeling for various data analysis tasks

Textbooks:

1. "An Introduction to Statistical Methods and Data analysis", R. Lyman Ott, Michael Longnecker, Sixth Edition, Cengage learning, 2010.
2. "Discrete-Event System Simulation", Jerry Banks, John S. Carson II, Barry L. Nelson, Fifth edition, Pearson Education.

Reference Books:

1. Lawrence M. Leemis, Stephen K. Park: Discrete – Event Simulation: A First Course, Pearson Education, 2006.
2. Averil M. Law: Simulation Modeling and Analysis, 4 th Edition, Tata McGrawHill, 2007.

KNOWLEDGE REPRESENTATION AND REASONING (PROFESSIONAL ELECTIVE I)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-I	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Introduction to Artificial Intelligence

Objectives: The Students will learn the following:

1. Planning and Game Playing for finding a procedural course of action for a declaratively described system to reach its goals while optimizing overall performance measures.
2. Constraint satisfaction method, Advanced Planning Method and Knowledge based Reasoning to separate the overlapping concerns with representations and reasoning
3. Logic, Inferences to build the mathematical knowledge for symbolic representation of propositions
4. Structured Knowledge Representations for symbolic representation of declarative knowledge.
5. Reasoning under Uncertainty to determine the use of intelligent agents to make decisions

Unit I: (09 Lectures)

Planning: STRIPS Domain, Forward State Space Planning, Backwards State Space Planning, Goal Stack Planning, Plan Space Planning, A Unified Framework for Planning

Game Playing: Board Games, Games Playing Algorithms, Limitations of Search, Other Games, Beyond Search.

Unit II: (10 Lectures)

Constraint Satisfaction Problem: N-Queens, Constraint Propagation, Scene Labelling, Higher Order Consistency, Directional Consistency, Algorithm Backtracking, Lookahead Strategies, Strategic Retreat.

Advanced Planning Methods: Graph Plan, Planning as Constraint Satisfaction, Planning as Satisfiability, Heuristic Search, Durative Actions, Trajectory Constraints and Preferences, Planning in the Real World.

Knowledge Based Reasoning: Agents, Facets of Knowledge

Unit III: (10 Lectures)

Logic and Inferences: Formal Logic, Propositional Logic, Resolution Method In Propositional Logic, First Order Logic, Incompleteness of Forward Chaining, Resolution Refutation in FOL, Deductive Retrieval, Complexity of Resolution Method in FOL. Horn Clauses and SLD Resolution, Backward Chaining, Second Order Logic.

Concepts and Language: Conceptual Domain, Reification, RDF and Semantic Web, Conceptual Dependency Theory

Unit IV: (10 Lectures)

Structured Knowledge Representations: Hierarchies in the Domain, The Schema, Frames, The Semantic Net, Scripts, Goals, Plans and MOPs, Inheritance in Taxonomies, Description Logics, Formal Concept Analysis, Conceptual Graphs.

Unit V: (09 Lectures)

Memory and Experience: Case Based Reasoning, Retrieval, Reuse and Adaptation

Reasoning Under Uncertainty: Default Reasoning, Qualitative Reasoning, Model based Diagnosis, Assumption based Reasoning and Truth Maintenance, Probabilistic Reasoning, Stochastic Actions, Combining Evidences to Form Beliefs

Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply Planning and Game Playing techniques to solve searching and to generate responsive and intelligent behaviors
2. Apply constraint satisfaction techniques and knowledge based reasoning to determine a solution to a set of constraints
3. Apply Logic and Inferences to generating the conclusions from evidence and facts and apply Concepts and Language to represent knowledge acquired from natural language input.
4. Analyze Structured Knowledge Representations for symbolic representation of declarative knowledge.
5. Apply Reasoning under Uncertainty to determine the use of intelligent agents to make decisions.

Textbooks:

1. A First Course in Artificial Intelligence, Deepak Khemani, Tata Mc-Graw Hill Education, 2013.

References:

1. AI – A Modern Approach, Stuart Russel, Peter Norvig, 4th Edition, Pearson Education, 2009.
2. Artificial Intelligence, Kevin Knight, Elaine Rich, Shivshankar B. Nair, 3rd Edition, Tata Mc-Graw Hill Education, 2019.

EXPERT SYSTEMS
(PROFESSIONAL ELECTIVE II)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-II	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Probability and Statistics

Course Objectives:

1. To study the idea of intelligent agents and expert systems.
2. To study about representing knowledge.
3. To study the reasoning and decision making in uncertain world.
4. To construct plans and methods for generating knowledge.
5. To study the concepts of expert systems tools and Applications.

Unit-I:

(10 Lectures)

Introduction to Expert Systems: Definition and Brief history of Expert systems, Features of an expert system, Organization of Expert System, Characteristics, Prospector, Knowledge Representation in expert systems.

Unit-II:

(08 Lectures)

Knowledge And Reasoning: Adversarial search, Optimal and imperfect decisions ,Alpha, Beta pruning, Logical agents: Propositional logic, First order logic, Syntax and semantics , Using first order logic, Inference in first order logic.

Unit-III:

(08 Lectures)

Uncertain Knowledge and Reasoning: Uncertainty, Acting under uncertainty, Basic probability notation , Axioms of probability, Baye's rule – Probabilistic reasoning ,Making simple decisions.

Unit-IV:

(12 Lectures)

Planning And Learning: Planning: Planning problem – Partial order planning – Planning and acting in non-deterministic domains – Learning: Learning decision trees – Knowledge in learning – Neural networks – Reinforcement learning – Passive and active.

Unit-V:

(12 Lectures)

Expert Systems Tools & Applications: Skeletal System- MYCIN, EXPERT, General Purpose System-OPS5, ROSIE, Applications of Expert systems.

Course Outcomes:

1. Interpret the concept of intelligent agents and expert systems.
2. Study about representing knowledge.
3. Apply reasoning and decision making in uncertain world.
4. Construct plans and methods for generating knowledge.
5. Analyse and apply concepts of expert systems tools and Applications.

Text Books:

1. Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education.
2. Stuart Russel and Peter Norvig, 'Artificial Intelligence A Modern Approach', Second Edition, Pearson Education, 2003 / PHI.

Reference Books:

1. George F.Luger, 'Artificial Intelligence – Structures and Strategies for Complex Problem Solving', 4th Edition, Pearson Education, 2002.
2. Elain Rich and Kevin Knight, 'Artificial Intelligence', Second Edition Tata McGraw Hill, 1999.

GAME PROGRAMMING
(PROFESSIONAL ELECTIVE II)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-II	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Discrete Mathematics

Course Objectives:

1. Understand the concepts of Game design and development.
2. Learn the processes, mechanics and issues in Game Design.
3. Be exposed to the Core architectures of Game Programming.
4. Know about Game programming platforms, and frame works.
5. Learn to develop 2d Scene games.

Unit-I:

(10 Lectures)

3D Graphics For Game Programming: 3D Transformations, Quaternions, 3D Modeling and Rendering, Ray Tracing, Shader Models, Lighting, Color, Texturing, Camera and Projections, Culling and Clipping, Character Animation, Physics-based Simulation, Scene Graphs.

Unit-II:

(08 Lectures)

Game Engine Design: Game engine architecture, Engine support systems, Resources and File systems, Game loop and real-time simulation, Human Interface devices, Collision and rigid body dynamics, Game profiling.

Unit-III:

(08 Lectures)

Game Programming: Application layer, Game logic, Game views, managing memory, controlling the main loop, loading and caching game data, User Interface management, Game event management.

Unit-IV:

(12 Lectures)

Building A Game System Framework: Real-Time Systems, The basic Game Loop, Using a KeyListener on a Frame and requesting the focus, Pseudo asynchronous input via the keyboard, Player Controlled Actions, Static Image based Sprites, Game Object Movement based on User Input, A.I. and Classical Mechanics.

Unit-V:

(12 Lectures)

2D Scene Building: Building an Animation Object for Frame based Animation, Building Animated Sprites, Translation in 2D space for Sprites, User and Computer control of Sprites, 2D Polygon, Models Building an Object to encapsulate 2D Polygon Models, Translation in 2D space revisited for 2D Polygon Models. Rotation in 2D space for 2D Polygon Models, User and Computer control of 2D Polygon Models, Backgrounds Scrolling and Parallax Scrolling.

Course Outcomes:

1. Infer the concepts of Game design and development.
2. Design the processes, and use mechanics for game development.
3. Relate the Core architectures of Game Programming.
4. Use Game programming platforms, and frame works.
5. Create interactive Games using 2D.

Text Books:

1. Mike Mc Shaffrly and David Graham, "Game Coding Complete", Fourth Edition, Cengage Learning, PTR, 2012.

Reference Books:

1. Jason Gregory, "Game Engine Architecture", CRC Press / A K Peters, 2009.
2. David H. Eberly, "3D Game Engine Design, Second Edition: A Practical Approach to Real-Time Computer Graphics" 2nd Edition, Morgan Kaufmann, 2006.
3. Developing Games in Java by David Brakeen, Core Techniques and Algorithms in Game Programming by Daniel Sanchez-Crespo Dalmau, and AI Techniques for Game Programming by Matt Buckland
4. Ernest Adams and Andrew Rollings, "Fundamentals of Game Design", 2nd Edition Prentice Hall / New Riders, 2009.

**COMPUTATIONAL LINGUISTICS
(PROFESSIONAL ELECTIVE II)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-II	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Probability and Statistics

Course Objectives:

1. Morphemes and use transducers for Modeling the Phonemes, Pronunciation Modelling.
2. N-Gram Model, Context Free Grammars, Parts of Speech Tagging, Syntax Analysis
3. Probabilistic CYK, Parsing of CFG, Unification of Feature Selection, Data Structures, Unification Algorithm
4. Semantic Analysis for Word Sense Disambiguation and Lexical Semantics.
5. Pragmatics of Language and Natural Language Generation for Text.

Unit-I:

(10 Lectures)

Introduction: Words, Regular Expressions and Automata, Morphology and Finite, State Transducers, Computational Phonology and Pronunciation Modeling, Probabilistic Models of Pronunciation and Spelling.

Unit-II:

(08 Lectures)

Syntax: N-gram models of Syntax, Word Classes and Parts of-Speech Tagging, Context-Free Grammars for English, Parsing with Context-Free Grammars.

Unit-III:

(08 Lectures)

Probabilistic Context-Free Grammars: Probabilistic CYK Parsing of PCFGs-Learning PCFG Probabilities-Problems with PCFGs-Probabilistic Lexicalized CFGs-Dependency Grammars-Human Parsing.

Unification of Feature Structures: Feature Structures in the Grammar, Agreement-Head Features, Subcategorization Long Distance Dependencies, Implementing Unification, Unification Data Structures, The Unification Algorithm.

Unit-IV:

(12 Lectures)

Semantics: Representing Meaning, Semantic Analysis, Lexical Semantics, Word Sense Disambiguation and Information Retrieval.

Unit-V:

(12 Lectures)

Pragmatics: Discourse, Dialog and Conversational Agents, Natural Language Generation, Machine Translation.

Course Outcomes:

1. Identify the Morphemes and use transducers for Modeling the Phonemes.
2. Apply the N-Gram Model to Context Free Grammars to obtain the resulting Parts of Speech Tagging, Unification of Feature Selection and Construct the Long Distance Dependencies.
3. Apply the CYK Algorithm to CFG to perform the Probabilistic Parsing
4. Identify the Lexical Semantics of the Phrases and Sentences for WSD.
5. Explain the features of Dialog and Conversational Agents, Machine Translation.

Text Books:

1. Speech and Language Processing - An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, 2nd Edition, Pearson Education.

Reference Books:

1. Statistical Language Learning, Charniak E, MIT Press.
2. Natural Language Understanding, Allen J, Cummins.

**PREDICTIVE ANALYTICS
(PROFESSIONAL ELECTIVE II)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-II	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Probability and Statistics

Course Objectives:

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics for Business
3. To introduce the tools, technologies & programming languages which is used in day to day analytics cycle.

Unit-I: (10 Lectures)

Introduction to Predictive Analytics & Linear Regression (NOS 2101): What and Why Analytics, Introduction to Tools and Environment, Application of Modelling, in Business, Databases & Types of data and variables, Data Modelling Techniques, Missing imputations. Need for Business Modelling, Regression – Concepts, Blue property-assumptions-Least Square Estimation, Variable Rationalization, and Model Building.

Unit-II: (08 Lectures)

Logistic Regression (NOS 2101): Model Theory, Model fit Statistics, Model Conclusion, Analytics applications to various Business Domains, Regression Vs Segmentation – Supervised and Unsupervised Learning, Tree Building – Regression, Classification, Over fitting, Pruning and complexity, Multiple Decision Trees.

Unit-III: (08 Lectures)

Objective Segmentation (NOS 2101): Regression Vs Segmentation – Supervised and Unsupervised Learning, Tree Building – Regression, Classification, Over fitting, Pruning and complexity, Multiple Decision Trees.

Develop Knowledge, Skill, and Competences (NOS 9005): Introduction to Knowledge skills & competences, Training & Development, Learning & Development, Policies and Record keeping.

Unit-IV: (12 Lectures)

Time Series Methods /Forecasting, Feature Extraction (NOS 2101): Arima, Measures of Forecast Accuracy, STL approach, Extract features from generated model as Height, Average, Energy etc and Analyze for prediction.

Unit-V: (12 Lectures)

Working with Documents (NOS 0703): Standard Operating Procedures for documentation and knowledge sharing, defining purpose and scope documents, Understanding structure of documents – case studies, articles, white papers, technical reports.

Course Outcomes:

1. Relate to basic concepts of Predictive analysis and linear algebra.
2. Infer logistic regression.
3. Relate object segmentation and develop knowledge.
4. Interpret time series methods and feature extraction.
5. Infer the working with documents.

Text Books:

1. Student's Handbook for Associate Analytics - III.

Reference Books:

1. An Introduction to Statistical Learning with Applications in R, Gareth James, Daniela Witten, Trevor Hastie Robert Tibshirani.

BLOCKCHAIN TECHNOLOGIES
(PROFESSIONAL ELECTIVE II)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-II	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Prerequisites: Data Structures, Information Security

Course Objectives:

1. To understand the fundamental concepts of blockchain and Bitcoin.
2. To learn about the transaction of bitcoins.
3. To understand the need for hashing and its working mechanisms.
4. To learn how to protect data stores and verify transactions.
5. To inculcate how to apply blockchain in real-time and the challenges faced.

Unit I: (10 Lectures)

Introduction: Bitcoin, History of Bitcoin, Bitcoin Users and Their Stories, Bitcoin Wallet, Structure of Blockchains, Types of Blockchain, Consensus, Blockchain Applications, Decentralization using blockchain, methods of decentralization, smart contracts, decentralization organization. (T1 Ch1)

Unit II: (10 Lectures)

How Bitcoin Works: Transactions, Blocks, Mining and the Blockchain, Bitcoin Overview, Bitcoin Transactions, constructing a Transaction, Bitcoin Mining, Mining Transactions in Blocks, Spending the Transaction. (R1 Ch2)

Unit III: (10 Lectures)

Blockchain Needed: Discovering the core problem, Disambiguating the Term, Understanding the Nature of Ownership, and Spending Money Twice.

How Blockchain Works: Part-1: Planning the blockchain, Documenting Ownership, Hashing Data, Hashing in the Real World, identifying and protecting user accounts. (T2 Ch2, Ch3)

Unit IV: (10 Lectures)

How Blockchain Works: Part-2: Authorizing Transactions, Storing Transaction Data, Using the Data Store, Protecting the Data Store, distributing the Data Store among peers, Verifying and Adding Transactions, choosing a Transaction History, and paying for integrity. (T2 Ch3)

Unit V: (08 Lectures)

Challenges: Technical Challenges, Market, Business Challenges, Legal, Regulatory Barriers and Behavioral, Educational Challenges, Blockchain Applications in Financial Services. Case study: Blockchain ensures instant claims processing: Etherisc. (R2 Ch3, Ch4)

Course Outcomes:

1. Understand the fundamental concepts of Blockchain and Bitcoins (L2)
2. Determine the working of Bitcoin. (L3)
3. Describe the need for blockchain and apply the hashing technique in real-time (L2, L3)
4. Distinguishing different techniques on blockchain to verify and add transactions (L4)
5. Analyze and implementation of Blockchain in financial services (L5, L6).

Text Books:

1. Blockchain For Dummies, Tiana Laurence, John Wiley & Sons publication, 2017, ISBN (Print) 978-1-119-36559-4.
2. Blockchain Basics: A Non-Technical Introduction in 25 Steps, Daniel Drescher, Apress publication, 2017, ISBN (print): 978-1-4842-2603-2.

Reference Books:

1. Mastering Bitcoin Unlocking Digital Crypto currencies, Andreas M. Antonopoulos, O'Reilly Publication, 1st Edition, Dec'-2014, ISBN: 978-1-449-37404-4.
2. The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology, William Mougayar, John Wiley & Sons publication, 2016, ISBN (print) 978-1-119-30031-1.

DEEP LEARNING LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The students will learn the following:

7. Make use of Data sets in implementing the machine learning algorithms
8. Implement the machine learning concepts and algorithms in Python language.

Week 1:

- i. Implement the model of a Neuron, The hidden-to-hidden layer, the hidden-to-output layer.

Week 2:

- i. Setup of TensorFlow, TensorFlow with GPU support.
- ii. Setup to Keras.
- iii. Learning the other deep learning libraries like Caffe, Theano, etc.

Week 3:

- i. Preparing the Data
- ii. Binary data and binary classification
- iii. Categorical data and multiple classes
- iv. Real-valued data and univariate regression

Week 4:

- i. Implement a program to training a single neuron.
- ii. Implement a program perceptron model.
- iii. Implement a program for perceptron learning algorithm (PLA)

Week 5:

- i. Implement a program for multiple layers on neurons.

Week 6:

- ii. Implement a program Wide neural networks
- iii. Implement a program Dense deep neural networks
- iv. Implement a program Sparse deep neural networks

Week 7:

- i. Implement a program for Convolutional Neural Networks
 - a. Loading data
 - b. Compiling the model
 - c. Training the CNN, and get the Results
 - d. Visualization of filters

Week 8:

- i. Implement the program for Recurrent Neural Networks

Week 9:

- i. Implement the program for Long short-term memory (LSTM) model

Week 10:

- i. Implement the program for Bi- directional Long short-term memory model.

Week 11:

1. Implement the program for Training a GAN
 - a. An MLP model
 - b. A convolutional model

Week 12:

Write a program exploring autoencoders.

Textbooks:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville, MIT Press.
2. Deep Learning from Scratch: Building with Python from First Principles, Seth Weidman, O'Reilly.

References:

1. Deep Learning for Beginners: A beginner's guide to getting up and running with deep learning from scratch using Python, Pablo Rivas, Packt Publishing Ltd, 2020.
2. Deep Learning with Tensorflow, Giancarlo Zaccone, Md. Rezaul Karim, Ahmed Menshawy,

DATA MINING LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Prerequisites: Probability and Statistics, Database Management Systems

Course Objectives:

1. To learn introductory and necessity of Data Mining and Data Warehousing.
2. To understand the importance of data pre-processing for data analysis.
3. To learn in depth about the association rules and to understand frequent item sets
4. Ability to classify the data into various formats and predicting the unknown data
5. To learn grouping of data objects without knowing class labels in prior.

List of Experiments:

1. Create a data warehouse on any dataset using OLAP Cube.
2. Develop Star, Snowflake and Fact constellation schemas using OLAP Cube.
3. Apply OLAP operations like Slice, Dice, Pivot, Select and Project etc.
4. Demonstration of preprocessing on dataset student.arff
5. Demonstration of preprocessing on dataset labor.arff
6. Demonstration of Association rule process on dataset contactlenses.arff using apriori algorithm
7. Demonstration of Association rule process on dataset test.arff using apriori algorithm
8. Demonstration of classification rule process on dataset student.arff using j48 algorithm
9. Demonstration of classification rule process on dataset employee.arff using j48 algorithm
10. Demonstration of classification rule process on dataset employee.arff using id3 algorithm
11. Demonstration of classification rule process on dataset employee.arff using naïve bayes algorithm
12. Demonstration of clustering rule process on dataset iris.arff using simple k-means
13. Demonstration of clustering rule process on dataset student.arff using simple kmeans.

Textbooks:

1. Data Mining Concepts and Techniques, Jiawei Han & Micheline Kamber, Morgan Kaufmann Publishers.
2. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Pearson.

Reference Books:

1. Data Mining Introductory and Advanced Topics, Margaret H Dunham, Pearson Education.
2. Data Mining Techniques, Arun K Pujari, University Press.
3. Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition.

III YEAR II SEM DATA ANALYTICS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. Understand the data science concepts and terminologies.
2. Interpret the data using basic statistics and data structures.
3. Explore data and analyze the data by using R functions and Packages.
4. Implementation of classification, clustering and feature selection methods with R.
5. Model the data to analyze and to predict using regression techniques.

Unit I:

(09 Lectures)

Introduction: Overview of Data Analytics, Need of Data Analytics, Characteristics of Data, Classification of Data: Structured, Semi-Structured, Unstructured, Business Intelligence, Data Analytics for Business Analysis. Overview of R programming, Environment setup with R Studio, R Commands, Variables and Data Types.

Unit II:

(12 Lectures)

Control Structures and Statistics: R Control Structures, Array, Matrix, Vectors, Factors, Functions, R packages, Probability Distributions, Normal Distribution- Binomial Distribution- Poisson Distributions Other Distributions, Basic Statistics, Correlation and Covariance, T-Tests, Hypothesis Testing and ANOVA.

Unit III:

(10 Lectures)

Exploratory Data Analysis: Data Visualization using R, Reading and getting data into R (External Data): Using CSV files, XML files, Web Data, JSON files, Databases, Excel files. Working with R Charts and Graphs: Histograms, Boxplots, Bar Charts, Line Graphs, Scatterplots, Pie Charts.

Unit IV:

(10 Lectures)

Classification and Clustering Techniques: Overview of Classification, Decision Trees, Decision Tree Algorithms, Decision Tree in R, Bayes Theorem, Naïve Bayes Classifier, Smoothing, Naïve Bayes in R, Random Forests. Overview of Clustering, K-means, Use Cases, Overview of the Method, perform a K-means Analysis using R.

Unit V:

(09 Lectures)

Regression Techniques: Linear Models, Simple Linear Regression, -Multiple Regression Generalized Linear Models, Logistic Regression, - Poisson Regression- other Generalized Linear Models-Survival Analysis, Nonlinear Models, Splines etc. and Case studies.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Understand the important keywords of Data Analytics and also learn how R Programming can play an important role in solving complex analytical problems.
2. Apply the knowledge of probability and statistics to test the significance of the data to be analyzed.
3. Identify the exploratory data analysis is an important step in the data analytics and also to work with different data formats.
4. Distinguish in between supervised and unsupervised machine learning algorithms for data modeling.
5. Analyze the data models to predict the future outcomes.

Textbooks:

1. An Introduction to R, Notes on R: A Programming Environment for Data Analysis and Graphics. W. N. Venables, D.M. Smith and the R Development Core Team. Version 3.0.1 (2013-05-16). URL: <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>
2. David Dietrich, Barry Heller and Beibei Yang, "Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data", EMC Education Services, Reprint 2015, Wiley, ISBN: 9788126556533.

References:

1. Big Data-Principles and best practices of scalable real-time data systems, R in Action, Rob Kabacoff, Manning Nathan Marz, James Warren, 2015, DreamTech Press. ISBN:9789351198062
2. The art of R programming: tour of statistical software design, Norman Matloff. ISBN-13: 978-1-59327-384-2.

NATURAL LANGUAGE PROCESSING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

- To understand the algorithms available for processing of information in Linguistics and Computational Properties of Natural Languages.
- To conceive basic knowledge on various morphological, syntactic and semantic NLP tasks.
- To familiarize various NLP software libraries and data sets publicly available.
- To develop systems for various NLP problems with moderate complexity.
- To learn various strategies for NLP system evaluation and error analysis.

Unit I:**(10 Lectures)**

Introduction to Natural Language Processing: The study of Language, Applications of NLP, Evaluating Language Understanding Systems, Different levels of Language Analysis, NLP Phases, Difficulty of NLP including ambiguity; Detecting and Correcting Spelling Errors and Noisy Channel Model; Minimum Edit Distance, Formal Grammar of Natural Grammar of English.

Unit II:**(10 Lectures)**

Language Modelling: Modeling with N-gram, Simple N-gram models, smoothing techniques, Evaluating language models; Neural network basics, Training; Neural Language Model, Case Study: application of Natural language model in NLP system development

Unit III:**(10 Lectures)**

Parts-of-Speech Tagging: Parts-of-speech Tagging: basic concepts; Tagset; Early approaches: Rule based and TBL; POS tagging using HMM, Introduction to POS Tagging using Neural Model.

Unit IV:**(08 Lectures)**

Parsing: Grammars and sentence Structure, Top-Down and Bottom-Up Parsers, Transition Network Grammars, Top- Down Chart Parsing. Syntactic parsing: CKY parsing; Statistical Parsing basics: Probabilistic Context Free Grammar (PCFG); Probabilistic CKY Parsing of PCFGs.

Unit V:**(08 Lectures)**

Semantics: Vector Semantics; Words and Vector; Measuring Similarity; Semantics with dense vectors; SVD and Latent Semantic Analysis; Skip-gram and CBOW; Concept of Word Sense; Introduction to WordNet.

Outcomes: Upon the successful completion of the course, the student will be able:

- Explain the concepts of morphology, syntax, semantics, discourse & pragmatics of natural language.
- Demonstrate understanding of the relationship between NLP and statistics & machine learning.
- Discover various linguistic and statistical features relevant to the basic NLP task, namely, spelling correction, morphological analysis, parts-of-speech tagging, parsing and semantic analysis.
- Interpret the usage of systems for various NLP problems with moderate complexity.
- Analyze the NLP systems, identify shortcomings and suggest solutions for these shortcomings.

Textbooks:

- Speech and Language Processing, Daniel Jurafsky, James H.Martin, 2nd Edition, Prentice Hall, 2008.

References:

- Speech and Language Processing - An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, 2nd Edition, Pearson Education.
- Statistical Language Learning, Charniak E, MIT Press.
- A Primer on Neural Network Models for Natural Language Processing, Goldberg Yoav.

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. Understand the financial statements and be able to analyze the financial statements.
2. Make the engineers familiar with the concepts of economics in business decisions, understand the techniques of capital budgeting.
3. Analyze demand and Production for price determinations.

Unit I:**(10 Lectures)**

Accounting for Engineers: Principles of Accounting, Final Accounts – Structure of Profit and Loss Account - Balance Sheet (Simple problems, No adjustments), Financial ratios – Importance and limitations of ratio analysis, Computation, Analysis and Interpretation of : Liquidity Ratios, Activity Ratios (Inventory Turnover ratio and Debtors Turnover ratio), Capital structure Ratios and Profitability ratios (Gross Profit Ratio, Net Profit Ratio & Operating Ratio) (simple problems).

Unit II:**(10 Lectures)**

Capital Budgeting: Nature and scope of capital budgeting, Methods of Capital Budgeting: Payback Method, Accounting Rate of Return (ARR) and Net Present Value Method (simple problems).

Unit III:**(10 Lectures)**

Introduction to Business Economics: Business economics meaning, Scope and objectives of Business economics, Nature and types of business decisions, Role and responsibilities of business economist.

Unit IV:**(08 Lectures)**

Introduction & Demand Analysis: Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand- Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting: methods of demand forecasting (survey methods, statistical methods, expert opinion method, test marketing, controlled experiments, judgmental approach).

Unit V:**(08 Lectures)**

Introduction to Production, Cost and Pricing: Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Laws of Returns, Internal and External Economies of Scale. Cost concepts. Pricing objectives- Methods of Pricing - Cost Plus Pricing, Marginal Cost Pricing, Sealed Bid Pricing, Going Rate Pricing, Limit Pricing, Market Skimming Pricing, Penetration Pricing, Two-Part Pricing, Block Pricing, Bundling Pricing, Peak Load Pricing, Cross Subsidization.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Understand the Accounting for Engineers
2. Analyse the Projects through Capital Budgeting Methods
3. Understand the Business Economics
4. Analyse the Demand through methods of Demand Forecasting
5. Evaluate Production and Pricing methods.

Textbooks:

1. Managerial Economics In a Global Economy, Domnick Salvatore, 4th Edition, Thomson.
2. Managerial Economics and Financial Analysis, Aryasri, 2nd Edition, TMH, 2005.

References:

1. General Economics, Deepashree, Tata Mcgrawhill
2. Business Economics, HL Ahuja, S.Chand
3. Financial Accounting, S.N.Maheswari, S.K. Maheswari, Vikas Publications.

**ENSEMBLE LEARNING
(PROFESSIONAL ELECTIVE III)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-III	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. To understand the need of ensemble methods.
2. To understand the Boosting and Bagging approaches.
3. To understand combination methods and diversity.
4. To understand the ensemble pruning and clustering ensembles.

Unit I: (08 Lectures)

Introduction: Basic concepts, popular learning algorithms: linear discriminant analysis, decision trees, neural networks, naïve bayes classifier, k-Nearest Neighbor, SVM and kernel methods, evaluation and comparison, ensemble methods, applications of ensemble methods.

Unit II: (10 Lectures)

Boosting: boosting procedure, the AdaBoost algorithm, examples, theoretical issues, multiclass extension, noise tolerance.

Bagging: two ensemble paradigms, the bagging algorithm, examples, theoretical issues, Random Tree ensembles: Random Forest, spectrum of randomizations, Random Tree ensembles for density estimation and anomaly detection.

Unit III: (08 Lectures)

Combination Methods: Benefits of Combination, averaging: simple and weighted. Voting: majority voting, plurality voting, weighted voting, soft voting, theoretical issues. Combining by learning: stacking, infinite ensemble. Other combination methods: algebraic methods, behavior knowledge space methods, decision template method. Relevant methods: error-correcting output codes, dynamic classifier selection, mixture of experts.

Unit IV: (08 Lectures)

Diversity: Ensemble diversity, Error decomposition, Diversity measures: pairwise, non-pairwise, summary and visualization, limitation of diversity measures. Information theoretic diversity: information theory and ensemble, interaction information diversity, multi-information diversity, estimation method. Diversity generation.

Unit V: (10 Lectures)

Ensemble Pruning: what is ensemble pruning, many could be better than all, categorization of pruning methods: ordering based, clustering based, optimization based pruning methods.

Clustering Ensembles: Clustering: methods, evaluation, need of clustering ensembles. Categorization of clustering ensemble methods, similarity based, graph-based, relabeling based, transformation-based methods.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Recall the basic concepts of ensemble learning
2. Infer Boosting and Bagging
3. Relate combination methods
4. Interpret diversity of ensemble methods
5. Infer ensemble pruning and clustering ensembles.

Textbooks:

1. Zhou, Zhi-Hua. Ensemble Methods: Foundations and Algorithms. United Kingdom, Taylor & Francis, 2012.

References:

1. Natarajan, Vijayalakshmi, and Sarkar, Dipayan. Ensemble Machine Learning Cookbook: Over 35 Practical Recipes to Explore Ensemble Machine Learning Techniques Using Python. India, Packt Publishing, 2019.
2. Rokach, Lior. Ensemble Learning: Pattern Classification Using Ensemble Methods (Second Edition). Singapore, World Scientific Publishing Company, 2019.

VIDEO ANALYTICS
(PROFESSIONAL ELECTIVE III)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-III	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Linear Algebra and Calculus

Objectives: The Students will learn the following:

1. To understand the need for video Analytics.
2. To understand the basic configuration of video analytics.
3. To understand the functional blocks of a video analytic system.
4. To get exposed to the various applications of video analytics.

Unit I:

(09 Lectures)

Components of Video Analytics: Video Analytics in Need-Video Analytics- Foreground extraction- Feature extraction of Video Analytics- Video Analytics classifier - Preprocessing- edge detection- Video Analytics smoothening- Feature space-PCA-FLD-SIFT features.

Unit II:

(09 Lectures)

Foreground In Extraction: Background estimation of Video Analytics- Averaging Video Analytics - Gaussian Mixture Model- Optical Flow based- Image Segmentation- Region growing- Region Splitting- Morphological operations- Erosion-Dilation- Tracking in a multiple camera environment.

Unit III:

(09 Lectures)

Classification: Introduction: Neural networks, Neural Networks (back propagation) – Various types of Deep learning networks- Fuzzy Classifier- Bayesian classifier-HMM based classifier

Unit IV:

(09 Lectures)

Security Based Video Analytics: Abandoned object detection- analysis of human behavior –recognition of Human - perimeter security- crowd analysis and prediction of crowd congestion.

Unit V:

(09 Lectures)

Video Analytics in Social Networks: Customer behavior analysis -people counting- Traffic rule violation detection- traffic congestion identification for route planning- driver assistance- lane change warning.

.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Infer the basics and the need of video analytics
2. Explain various methods to extract videos in foreground
3. Classify video analytic algorithms for security applications
4. Applying video analytic algorithms for business intelligence
5. Design custom made video analytics system for the various social networks.

Textbooks:

1. Graeme A. Jones (Editor), Nikos Paragios (Editor), Carlo S. Regazzoni (Editor) Video-Based Surveillance Systems: Computer Vision and Distributed Processing, Kluwer academic publisher, 2001.

References:

1. Nilanjan Dey (Editor), Amira Ashour (Editor) and Suvojit Acharjee (Editor), Applied Video Processing in Surveillance and Monitoring Systems (IGI global) 2016
2. Zhihao Chen (Author), Ye Yang (Author), Jingyu Xue (Author), Liping Ye (Author), Feng Guo (Author), The Next Generation of Video Surveillance and Video Analytics: The Unified Intelligent Video Analytics Suite, Create Space Independent Publishing Platform, 2014
3. Caifeng Shan (Editor), Fatih Porikli (Editor), Tao Xiang (Editor), Shaogang Gong (Editor) Video Analytics for Business Intelligence, Springer, 2012.

**APPLIED NATURAL LANGUAGE PROCESSING
(PROFESSIONAL ELECTIVE III)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-III	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Linear Algebra and Calculus

Objectives: The Students will learn the following:

1. Vector Sequence Model, Statistical Properties of Words
2. Vector Space Models, Representation of Words as Vectors
3. Language Models and Application of Neural Networks in NLP.
4. Softmax and its effect in Language Models, Sequence Learning.
5. Statistical and Neural Machine Translation.

Unit I: (10 Lectures)

Introduction to Natural Language Understanding: Introduction, Operations on a Corpus, Probability and NLP, Vector Sequence Model, Sequence Learning, Machine Translation, Preprocessing Statistical Properties of Words.

Unit II: (09 Lectures)

Word to Vectors: Vector Space Models for NLP, Document Similarity, Vector Representation of Words, Co-Occurrence Matrix, Query Processing, Topic Modeling.

Unit III: (10 Lectures)

Probability and Language Model: Introduction to Probability in the Context of NLP, Joint and Conditional Probabilities, Probabilistic Language Model, Chain Rule and Markov Assumption, Generative Models, N-Gram Models.

Neural Networks for NLP-I: Machine Learning, Perceptron, Linearly Separable, Linear Models for Classification, Biological Neural Network, Perceptron Learning, Activation Functions, Gradient Descent.

Unit IV: (09 Lectures)

Neural Networks for NLP-II: Feedforward and Backpropagation Neural Network, Word2Vec, CBOW and Skip Gram Models, Reduction of Sampling-Sub-sampling, Negative Sampling, Binary Tree, Hierarchical Softmax, Sequence Learning and Its Applications.

Unit V: (11 Lectures)

Statistical Machine Translation: Machine Translation, Phrase-based Translation, Summarization of Alignments, Extraction of Phrases, Evaluation of Machine Translation

Neural Machine Translation: Encoder-Decoder Model for Neural Machine Translation, RNN based Machine Translation, Attention based Translation, Conversion Modelling, IR-based Conversion Modelling.

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Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply the Operations of Corpus on WordNet and examine the statistical properties of words.
2. To use existing natural language processing tools to conduct basic natural language processing, such as text normalization, named entity extraction, or syntactic parsing.
3. Apply the Probabilistic Language Model to NLP and perform the Probabilistic Parsing and Apply the Neural Networks to Natural Language Processing.
4. Apply Neural Networks to WordNet and examine the performance of Sampling on WordNet.
5. Decomposing a real-world problem into sub problems in natural language processing and identify potential solutions in Machine Translation.

Textbooks:

1. Natural Language Understanding, Allen James, Pearson Education.
2. Applied Natural Language Processing with Python, Taweh Beysolow II, Apress.

References:

1. Speech and Language Processing - An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, 2nd Edition, Pearson Education.
2. Statistical Language Learning, Charniak E, MIT Press.

**BUSINESS INTELLIGENCE
(PROFESSIONAL ELECTIVE III)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-III	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Nil**Objectives:** The Students will learn the following:

1. To understand the concept of Business Intelligence
2. To know the details of Decision Support System
3. To inculcate the concepts of Data Warehousing
4. To understand the basics of design and management of BI systems.

Unit I:**(10 Lectures)**

An Overview of Business Intelligence: Introduction to data, Information and knowledge, Evolution of DSS into Business Intelligence A Framework for Business Intelligence (BI)-, Introduction to Business Intelligence, Determining BI Cycle, BI Environment and Architecture, Identify BI opportunities, Benefits of BI. Role of Mathematical model in BI, Factors Responsible for successful BI Project, Obstacle to Business Intelligence in an Organization

Unit II:**(10 Lectures)**

Decision Making Concepts: Concepts of Decision Making, Techniques of Decision Support System (DSS), Development of Decision Support System (DSS), Applications of DSS, Role of Business Intelligence in DSS.

Unit III:**(08 Lectures)**

Data-Warehouse: Introduction: Data warehouse Modeling, data warehouse design, data-ware-house technology, Distributed data warehouse, and materialized view

Unit IV:**(10 Lectures)**

Data Pre-processing and outliers: Data Analytics life cycle, Discovery, Data preparation, Pre-processing requirements, data cleaning, data integration, data reduction, data transformation, Data discretization, and concept hierarchy generation, Model Planning, Model building, Communicating Results and Findings, Operationalizing, Introduction to OLAP. Real-world Applications, types of outliers, outlier challenges, Outlier detection Methods, Proximity-Based Outlier analysis, Clustering Based Outlier analysis.

Unit V:**(08 Lectures)**

Designing and managing BI systems: Determining infrastructure requirements, planning for scalability and availability, managing and maintenance of BI systems, managing BI operations or business continuity.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Interpret the basic concepts of BI.
2. Infer the decision-making concepts
3. Relate the DW concepts.
4. Interpret the data pre-processing and outliers.
5. Relate the designing and managing BI systems.

Textbooks:

1. R. Sharda, D. Delen, and E. Turban, Business Intelligence and Analytics. Systems for Decision Support, 10th Edition. Pearson/Prentice Hall, 2015. ISBN-13: 978-0-13-305090-5, ISBN-10: 0-13-305090-4.

References:

1. Business Process Automation, Sanjay Mohapatra, PHI.
2. Introduction to business Intelligence and data warehousing, IBM, PHI, ISBN: 9788120339279.

DEVOPS
(PROFESSIONAL ELECTIVE III)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites:**Objectives:** The Students will learn the following:

1. To understand the DevOps Concepts and Tools.
2. To determine about Deployment of Main DevOps Tools.
3. To learn various Pipeline basics and Artifacts.
4. To determine about Continuous Delivery using Docker Containerization.
5. To learn various Deployment Tools using Ansible and AWS Cloud.

Unit I:**(9 Lectures)****Software Development Life Cycle(SDLC):** Introduction to SDLC, Agile Model.

Introduction to Devops: DevOps Features, Work Management, Source Code Management, Build Automation, Delivery Automation, Understanding Code Quality, Automation of CI/CD.

Unit II:**(10 Lectures)****Source Code Management (GIT):** What is Version Control and GIT, GIT Installation, Working with remote repository, Standard Branching Workflows, Branching Workflow – GitFlow.

Unit testing – code coverage: jUnit, NUnit & Code Coverage with Sonar Qube, SonarQube – Code Quality Analysis. Build Automation - CI: Build Automation, What is CI , Why CI is Required , Introduction to Jenkins (With Architecture), Build (CI) Orchestration using Jenkins Automation Server, Introduction to Build Tools – Apache Maven, Gradle, Ant, NPM/Node.js.

Unit III:**(10 Lectures)****Pipeline Basics:** Jenkins Master, Node, Agent, and Executor, Automation Server - Jenkins: Continuous Integration and Delivery Server.

Artifact Management: Nexus, JFrog Artifactory as Kubernetes registry, Helm Chart for Microsoft Azure Pipeline.

Unit IV:**(08 Lectures)****Continuous Delivery:** Software components in short cycles. Deploying to Dev Environment.

Continuous Deployment: CD Flow, Containerization with Docker: Introduction to Docker, Images & Containers, DockerFile, Working with containers and publish to Docker Hub

Unit V:**(10 Lectures)****Continuous Deployment:** Configuration Management – Ansible: Introduction to Ansible, Ansible tasks, Roles, Jinja2 templating, Vaults, Deployments using Ansible.**Containerization using Kubernetes:** Introduction to Kubernetes, Namespace & Resources, CI/CD – On OCP, BC , DC & ConfigMaps, Deploying Apps on Openshift, Container Pods.**Aws & Azure – Cloud:** Introduction to AWS & Azure Clouds, Pipeline of AWS & Azure Clouds – CI/CD**Outcomes:** Upon the successful completion of the course, the student will be able:

1. Explain the basic concepts of DevOps and DevOps Tools.
2. Demonstrate different versions of Source Code Management using Git.
3. Determine about Pipeline Basics and Artifact Management.
4. Build and Automate Test using Jenkins and Maven, Containerization using Docker.
5. Design Automation Tool Application using ANSIBLE and Deployment using Kubernetes.

Textbooks:

1. The DevOps Handbook:: How to Create World-Class Agility, Reliability, and y Gene Kim, Jez Humble, Patrick Debois, John Willis.
2. Practical DevOps By Joakim Verona.

References:

1. DevOps for Developers By Michael Huttermann.

**REINFORCEMENT LEARNING
(PROFESSIONAL ELECTIVE IV)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-IV	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Machine Learning

Objectives: The Students will learn the following:

1. Multi-Armed Bandits.
2. Finite Markov Decision Processes.
3. Dynamic Programming and Evaluation.
4. Monte Carlo Methods and Decision Process
5. Temporal Difference Learning and Policy Control

Unit I

(11 Lectures)

Introduction to Reinforcement Learning: Reinforcement Learning, Elements of Reinforcement Learning, Limitations and Scope.

Multi-Armed Bandits: A K-Armed Bandit Problem, Action-Value Methods, 10-Armed Testbed, Incremental Implementation, Tracking a Nonstationary Problem, Optimistic Initial Values, Upper Confidence Bound Action Selection, Gradient Bandit Algorithm, Associative Search.

Unit II

(09 Lectures)

Finite Markov Decision Processes: Agent-Environment Interface, Goals and Rewards, Returns and Episodes, Unified Notation for Episodic and Continuing Tasks, Policies and Value Functions, Optimal Policies and Optimal Value Functions, Optimality and Approximation.

Unit III

(10 Lectures)

Dynamic Programming: Policy Evaluation, Policy Improvement, Policy Iteration, Value Iteration, Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming.

Unit IV

(10 Lectures)

Monte Carlo Methods: Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control Without Exploring Starts, Off-Policy Prediction via Importance Sampling, Incremental Implementation, Off-Policy Monte Carlo Control, Discounting Aware Importance Sampling, Per-Decision Importance Sampling.

Unit V

(09 Lectures)

Temporal Difference Learning: Temporal Difference (TD) Prediction, Advantages of TD Prediction Models, Optimality of TD(0), Sarsa On-Policy TD Control, Q-Learning Off-Policy TD Control, Expected Sarsa, Maximization Bias and Double Learning, Games, Afterstates and Other Special Cases.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Summarize the Elements of Reinforcement Learning and Apply K-Armed Bandit Problem for Different Situations.
2. Apply Finite Markov Decision Process and model a Control Task in the Markov Decision Process Framework..
3. Apply Dynamic Programming for Policy Evaluation and Improvement.
4. Outline the Monte Carlo Methods and Sampling Techniques.
5. Apply Temporal Difference for Prediction Models.

Textbook:

1. Reinforcement Learning: An Introduction, Richard S Sutton, Andrew G Barto, Second Edition, MIT Press.

References:

1. Reinforcement Learning, Phil Winder, O'Reily.

**VIRTUAL AND AUGMENTED REALITY
(PROFESSIONAL ELECTIVE IV)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-IV	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Nil

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E.

Unit I: (10 Lectures)

Introduction of Virtual Reality: Fundamental Concept and Components of Virtual Reality. Primary Features and Present Development on Virtual Reality. The historical development of VR: Scientific landmarks Computer Graphics, Real-time computer graphics, Flight simulation, Virtual environments, Requirements for VR, benefits of Virtual reality.

Unit II: (10 Lectures)

Computer Graphics and Geometric Modelling: Introduction, The Virtual world space, positioning the virtual observer, the perspective projection, human vision, stereo perspective projection, Color theory, Conversion from 2D to 3D, 3D space curves, 3D boundary representation, Simple 3D modelling, 3D clipping, Illumination models, Reflection models, Shading algorithms.

Unit III: (10 Lectures)

Virtual Environment: Input: Tracker, Sensor, Digital Gloves, Movement Capture, Video-based Input, 3D Menus & 3D Scanner etc. Output: Visual /Auditory / Haptic Devices. **Generic VR system:** Introduction, Virtual environment, Computer environment, VR technology, Model of interaction, VR Systems.

Unit IV: (10 Lectures)

Augmented Reality: Taxonomy, technology and features of augmented reality, difference between AR and VR Challenges with AR, AR systems and functionality, Augmented reality methods, visualization techniques for augmented reality, enhancing interactivity in AR environments, evaluating AR systems.

Unit V: (08 Lectures)

Development Tools and Frameworks: Human factors: Introduction, the eye, the ear, the somatic senses. Hardware: Introduction, sensor hardware, Head-coupled displays, Acoustic hardware, Integrated VR systems. Software: Introduction, Modelling virtual world, Physical simulation, VR toolkits, Introduction to VRML.

Outcomes: Upon the successful completion of the course, the student will be able:

1. To understand fundamental computer vision, computer graphics and human-computer interaction techniques related to VR/AR.
2. To understand geometric modeling and Virtual environment.
3. To relate and differentiate VR/AR technology.
4. To use various types of Hardware and software in virtual Reality systems
5. To implement Virtual/Augmented Reality applications.

Textbooks:

1. Grigore C. Burdea, Philippe Coiffet. Virtual Reality Technology, Wiley 2016.

References:

1. Alan B. Craig, Understanding Augmented Reality, Concepts and Applications, Morgan Kaufmann, 2013.
2. Anand R., "Augmented and Virtual Reality", Khanna Publishing House, Delhi.

**SEQUENCE MODELS IN NLP
(PROFESSIONAL ELECTIVE IV)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-IV	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Nil**Objectives:** The Students will learn the following:

1. Train a neural network with GLoVe word embeddings to perform sentiment analysis of tweets;
2. Generate synthetic Shakespeare text using a Gated Recurrent Unit (GRU) language model;
3. Train a recurrent neural network to perform named entity recognition (NER) using LSTMs with linear layers; and
4. Use so-called 'Siamese' LSTM models to compare questions in a corpus and identify those that are worded differently but have the same meaning.

Unit I:**(10 Lectures)**

Vector Space Models: Use vector space models to discover relationships between words and use principal component analysis (PCA) to reduce the dimensionality of the vector space and visualize those relationships

Word Embeddings and Locality Sensitive Hashing for Machine Translation: Write a simple English-to-French translation algorithm using pre-computed word embeddings and locality sensitive hashing to relate words via approximate k-nearest neighbor's search

Unit II:**(10 Lectures)**

Sentiment with Neural Nets: Train a neural network with GLoVe word embeddings to perform sentiment analysis of tweets with real time applications

Unit III:**(08 Lectures)**

Language Generation Models: Generate synthetic Shakespeare text using a Gated Recurrent Unit (GRU) language model with real time applications

Unit IV:**(08 Lectures)**

Named Entity Recognition (NER): Train a recurrent neural network to perform NER using LSTMs with linear layers with real time applications

Unit V:**(10 Lectures)**

Siamese Networks: Siamese Networks, Pros and Cons of Siamese Networks, Loss functions used in Siamese Networks, collect sample dataset and preprocessing it, Training the network, Testing the model.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Outline the Natural Language Processing and Vector space models
2. Formulate Sentimental Analysis tweets with real time applications
3. Apply and Evaluate Gated Recurrent Model
4. Formulate the LSTM Model
5. Illustrate the Siamese Networks.

Textbooks:

1. Hirst, Graeme, and Goldberg, Yoav. Neural Network Methods in Natural Language Processing. United States, Morgan & Claypool Publishers, 2017.
2. Liu, Yang. Deep Learning in Natural Language Processing. Germany, Springer Singapore, 2018.

References:

1. Patel, Ankur A., and Arasanipalai, Ajay Uppili. Applied Natural Language Processing in the Enterprise. N.p., O'Reilly Media, 2021.
2. Vajjala, Sowmya, et al. Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems. Japan, O'Reilly, 2020.

**SOCIAL NETWORKS AND SEMANTIC WEB
(PROFESSIONAL ELECTIVE IV)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PE-IV	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Nil**Objectives:** The Students will learn the following:

1. To understand the importance of semantic web
2. To understand the role of ontology and inference engines in semantic web
3. To understand and apply ontology methods on various domains.
4. To differentiate the services provided by ontologies with web services.
5. To build semantic web applications with social network features.

Unit I:**(10 Lectures)**

Thinking and Intelligent Web Applications, The Information Age, The World Wide Web, Limitations of Today's Web, The Next Generation Web, Machine Intelligence, Artificial Intelligence, Ontology, Inference engines, Software Agents, Berners-Lee www, Semantic Road Map, Logic on the semantic Web

Unit II:**(10 Lectures)**

Ontologies and their role in the semantic web, Ontologies Languages for the Semantic Web –Resource Description Framework(RDF) / RDF Schema, Ontology Web Language(OWL), UML, XML/XML Schema.

Unit III:**(10 Lectures)**

Ontology Engineering, Constructing Ontology, Ontology Development Tools, Ontology Methods, Ontology Sharing and Merging, Ontology Libraries and Ontology Mapping.

Unit IV:**(10 Lectures)**

Logic, Rule and inference Engines. Semantic Web applications and services, Semantic Search, e-learning, Semantic Bioinformatics, Knowledge Base.

Unit V:**(08 Lectures)**

XML based Web Services, Creating an OWL-S ontology for Web Services, Semantic Search Technology, Web Search Agents and Semantic Methods, What is social Network analysis, development of the social networks analysis, Electronic Sources for Network Analysis- Electronic Discussion networks. Blogs and Online communities, Web Based Networks. Building Semantic Web Applications with social features.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Differentiating semantic web from others.
2. Using ontology and inference engines in semantic web development.
3. Using ontology development tools to create ontologies
4. Identify the ontology web services for better user understanding.
5. Build semantic web applications with social network features, development of the social networks analysis.

Textbooks:

1. Thinking on the Web - Berners Lee, Godel and Turing, Wiley inter science, 2008.
2. Social Networks and the Semantic Web, Peter Mika, Springer, 2007.

References:

1. Semantic Web Technologies, Trends and Research in Ontology Based Systems, J.Davies, R.Studer, P.Warren, John Wiley & Sons.
2. Semantic Web and Semantic Web Services -Liyang Lu Chapman and Hall/CRC
3. Publishers,(Taylor & Francis Group)
4. Information Sharing on the semantic Web - HeinerStuckenschmidt; Frank Van Harmelen, Springer Publications.
5. Programming the Semantic Web, T.Segaran, C.Evans, J.Taylor, O'Reilly, SPD.

**DATA MODELLING AND DESIGN
(PROFESSIONAL ELECTIVE IV)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-I		PE-IV	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. To understand the foundation concepts of Conceptual Data Model and ER Model Grammar.
2. To learn the basic knowledge of ER Modelling and EER Modelling concepts.
3. To learn various concepts of Logical Data Model Operations.
4. To learn Relational Data Model Concepts using Structured Query Language (SQL).
5. To learn various Data Manipulation Operations and OO Modelling concepts.

Unit I:

(10 Lectures)

Foundation Concepts -Conceptual Data Modelling: Foundation Concepts: A Conceptual Modelling Framework, ER Modelling Primitives, **Foundations of the ER Modelling Grammar:** Entity Types and Attributes, Entity and Attribute-Level Data Integrity Constraints, Relationship Types, Structural Constraints of a Relationship Type, Base Entity Types and Weak Entity Types, Data Modelling Errors.

Unit II:

(10 Lectures)

Entity-Relationship (ER) Modelling: Superclass/subclass Relationship: Vignette 1, A Motivating Exemplar, General Properties of a Superclass/subclass Relationship, Specialization and Generalization, Specialization Hierarchy and Specialization Lattice, Categorization, Choosing the Appropriate EER Construct, Aggregation.

Converting from the Presentation Layer to a Design-Specific EKR Diagram, ER Model for the Revised Story

Unit III:

(10 Lectures)

Logical Data Modelling: Definition, Characteristics of a Relation, Data Integrity Constraints: The Concept of Unique Identifiers, Referential Integrity Constraint in the Relational Data Model, A Brief Introduction to Relational Algebra: Unary Operations: Selection and Projection, Binary Operations: Union, Difference, and Intersection, The Natural Join (*) Operation.

Views and Materialized Views in the Relational Data Model, Mapping an ER Model to a Logical Schema: Information-Reducing Mapping of ER Constructs, An Information-Preserving Mapping.

Unit IV:

(08 Lectures)

Relational Data model: Data Definition using SQL, Data Population using SQL, Access Control in the SQL-92 Standard, Relational Algebra, Structured Query Language (SQL).

Unit V:

(08 Lectures)

Advanced Data Manipulation: Assertions. Triggers and Views: Specifying an Assertion in SQL, Triggers in SQL, Specifying Views in SQL/DDDL, The Division Operation, SQL-92 Built-in Functions: The SUBSTRING Function, The CHAR LENGTH (char) Function, The TRIM Function, The TRANSLATE Function, The POSITION Function, Combining the INSTR and SUBSTR Functions.

Object Oriented Data Modelling Architecture: The Object-Oriented Data Model: Overview of OO Concepts, A Note on UML, The Object-Relational Data Model.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Explain the importance of Conceptual Data Model and ER Modelling Grammar.
2. Interpret the ER Modelling into Database Design.
3. Classify various categories of constraints and operations available in Logical Data Model.
4. Interpret the usage of Relational Data Model into SQL Language.
5. Analyze the functions of Advanced Data Manipulation and OO Data Modelling Architecture.

Textbooks:

1. Data Modelling and Database Design, Richard Scamell, Narayan Umanath. Martin, 2nd Edition, Cengage Learning, 2014.
2. Database Modelling and Design, Toby J. Teorey, Sam. S. Lightstone, Tom Nadeau, H. V. Jagadish, 5th Edition, Morgan Kaufmann, 2011.

Reference Books:

1. Data Modeling Essentials, Graeme C. Simsion, Graham. C. Witt, 3rd Edition, Morgan Kaufmann.

DATA ANALYTICS LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Course Objectives: The students will learn the following

1. How to program in R and how to use R for effective data analysis.
2. Install and configure software necessary for a statistical programming environment.
3. Carry out basic data pre-processing and wrangling in R-Studio and to explore practical issues in statistical computing.
4. Read in data into the R environment from different sources
5. Implement supervised/unsupervised machine learning techniques

Recommended Systems/Software Requirements:

1. Download and install R from The Comprehensive R Archive Network
URL: <https://cran.r-project.org/bin/windows/base/>
2. Download and install RStudio from www.rstudio.com.

Week 1: Overview of R, R data types and objects, reading and writing data

Week 2: Control structures, functions, scoping rules, dates and times etc.

Week 3: Data Structures (vectors, arrays, matrices, data frames and lists)

Week 4: Working with CSV files, XML files, Web Data, JSON files, Databases, Excel files

Week 5: Data Analysis I (Descriptive Statistics)

Week 6: Data Analysis I (ANOVA and Correlation)

Week 7: Data Visualization in R (Basic Plots: Scatter, Bar, Line, Box, ...)

Week 8: Data Visualization in R (Histograms and Distributions)

Week 9: Data Analysis II (Clustering & Classification)

Week 10: Data Analysis II (Regression)

Week 11: Case Studies

- The Economist's Big Mac Price Index
- U.S. Stock Market: SP500 and VIX

Week 12: Case Studies

- Covid-19 Datasets
- Agricultural Datasets.

Course Outcomes: After completion of the course, the students would be able to

1. Infer the R Programming language conventions for data analysis
2. Interpret the functioning of datatypes, variables and control structures on R Platform
3. Assess the significance of data/ datasets by applying the statistical tests using R.
4. Evaluate the performance of machine learning models.
5. Create and edit visualizations with R

Textbooks:

1. Computerworld's Beginners' Guide to R, June 2013.
2. An Introduction to R, by W. N. Venables, D. M. Smith and the R Core Team, 2015.
3. An Introduction to R for Quantitative Economics: Graphing, Simulating and Computing, Vikram Dayal, Springer Briefs in Economics, Springer (India), 2015.
4. Applied Econometrics with R, Christian Kleiber and Achim Zeileis, Springer-Verlag, New York, 2008.

NATURAL LANGUAGE PROCESSING LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	III-II		PC	L	T	P	C	CIE	SEE	Total
				-	-	2	1	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. Features of NLTK Library and Data Visualization Technique to analyses the text.
2. Stemming, Lemmatization techniques and PoS Tagging with all values.
3. Chunking and Chinking process to extract and exclude parts from a Text file.
4. Infer the usage of Named Entity Recognition and WordNet Database.
5. Bag of Words methods and TF-IDF factor.

Recommended Systems/Software Requirements:

1. Download and install Python URL: <https://www.python.org/downloads/>
2. Download and install Anaconda Navigator for Jupyter Notebook or use any IDE.

Week 1: Exploring Features of NLTK Library:

- Open the text file for processing. Import required libraries.
- Sentence tokenizing. Word tokenizing.
- Listing stop words and removing stop words. Find the final frequency distribution.

Week 2: Implement the Word Cloud Data Visualization Technique to know the analysis of Text.

Week 3: Implement the Porter's Stemmer and Snowball Stemmer algorithm to normalize the word using Stemming Technique.

Week 4: Implement the working of Lemmatizer technique with default PoS value and different PoS values.

Week 5: Demonstration of PoS tagging with all PoS Tags.

Week 6: Implement the Chunking Process i.e., to extract noun phrases from a text file.

Week 7: Implement the Chinking Process i.e., to exclude some parts from the chunked text.

Week 8: Demonstrate the use of Named Entity Recognition (NER) with 2 options when binary value is True and when binary value is False.

Week 9: Demonstrate the use of WordNet Database to find different definitions of a word, Synonyms, Antonyms, Hypernyms, and Hyponyms.

Week 10: Implement the Bag-of-Words method to convert the raw text into words and counts the frequency for the words in the text.

Week 11: Implementation of TF-IDF Term Frequency – Inverse Document Frequency measure which is used for Information Retrieval and Summarization Process.

Week 12: Implement Twitter Sentiment Analysis using different NLP techniques.

Course Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply the NLTK Library features to do analysis of words and to design Data Visualization Technique.
2. Infer the usage of different Stemming and Lemmatization techniques to normalize the words from text with different PoS values.
3. Apply Named Entity Recognition System with binary valued options to find out whether a particular word is named entity or not.
4. Construct the final model for own Bag of Words with the combination of all given input words and compute the TF-IDF score to measure the frequency of words.

Textbooks:

1. Speech and Language Processing, Daniel Jurafsky and James H.Martin, Prentice Hall, 2nd Edition,2008.

References:

1. Natural Language Processing with Python, Steven Bird,Ewan Klein, Edward Loper, O'Reilly Media;1st Edition,2009.
2. Foundations of Computational Linguistics: Human Computer Communication in Natural Language, Roland R.Hausser, Paperback, MIT Press, 2011.
3. Statistical Language Learning, Charniak E, MIT Press.

**IV YEAR I SEM
COMPUTER VISION**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. Understand the data science concepts and terminologies.
2. Interpret the data using basic statistics and data structures.
3. Explore data and analyze the data by using R functions and Packages.
4. Implementation of classification, clustering and feature selection methods with R.
5. Model the data to analyze and to predict using regression techniques.

Unit I:

(09 Lectures)

Introduction: Overview of Data Analytics, Need of Data Analytics, Characteristics of Data, Classification of Data: Structured, Semi-Structured, Unstructured, Business Intelligence, Data Analytics for Business Analysis. Overview of R programming, Environment setup with R Studio, R Commands, Variables and Data Types.

Unit II:

(12 Lectures)

Control Structures and Statistics: R Control Structures, Array, Matrix, Vectors, Factors, Functions, R packages, Probability Distributions, Normal Distribution- Binomial Distribution- Poisson Distributions Other Distributions, Basic Statistics, Correlation and Covariance, T-Tests, Hypothesis Testing and ANOVA.

Unit III:

(10 Lectures)

Exploratory Data Analysis: Data Visualization using R, Reading and getting data into R (External Data): Using CSV files, XML files, Web Data, JSON files, Databases, Excel files. Working with R Charts and Graphs: Histograms, Boxplots, Bar Charts, Line Graphs, Scatterplots, Pie Charts.

Unit IV:

(10 Lectures)

Classification and Clustering Techniques: Overview of Classification, Decision Trees, Decision Tree Algorithms, Decision Tree in R, Bayes Theorem, Naïve Bayes Classifier, Smoothing, Naïve Bayes in R, Random Forests. Overview of Clustering, K-means, Use Cases, Overview of the Method, perform a K-means Analysis using R.

Unit V:

(09 Lectures)

Regression Techniques: Linear Models, Simple Linear Regression, -Multiple Regression Generalized Linear Models, Logistic Regression, - Poisson Regression- other Generalized Linear Models-Survival Analysis, Nonlinear Models, Splines etc. and Case studies.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Understand the important keywords of Data Analytics and also learn how R Programming can play an important role in solving complex analytical problems.
2. Apply the knowledge of probability and statistics to test the significance of the data to be analyzed.
3. Identify the exploratory data analysis is an important step in the data analytics and also to work with different data formats.
4. Distinguish in between supervised and unsupervised machine learning algorithms for data modeling.
5. Analyze the data models to predict the future outcomes.

Textbooks:

1. An Introduction to R, Notes on R: A Programming Environment for Data Analysis and Graphics. W. N. Venables, D.M. Smith and the R Development Core Team. Version 3.0.1 (2013-05-16). URL: <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>
2. David Dietrich, Barry Heller and Beibei Yang, "Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data", EMC Education Services, Reprint 2015, Wiley, ISBN: 9788126556533.

References:

1. Big Data-Principles and best practices of scalable real-time data systems, R in Action, Rob Kabacoff, Manning Nathan Marz, James Warren, 2015, DreamTech Press. ISBN:9789351198062

2. The art of R programming: tour of statistical software design, Norman Matloff. ISBN-13: 978-1-59327-384-2.

DATA VISUALIZATION TECHNIQUES

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Machine Learning

Objectives: The Students will learn the following:

1. Understand the data science concepts and terminologies.
2. Interpret the data using basic statistics and data structures.
3. Explore data and analyze the data by using R functions and Packages.
4. Implementation of classification, clustering and feature selection methods with R.
5. Model the data to analyze and to predict using regression techniques.

Unit I:**(09 Lectures)**

Introduction: Overview of Data Analytics, Need of Data Analytics, Characteristics of Data, Classification of Data: Structured, Semi-Structured, Unstructured, Business Intelligence, Data Analytics for Business Analysis. Overview of R programming, Environment setup with R Studio, R Commands, Variables and Data Types.

Unit II:**(12 Lectures)**

Control Structures and Statistics: R Control Structures, Array, Matrix, Vectors, Factors, Functions, R packages, Probability Distributions, Normal Distribution- Binomial Distribution- Poisson Distributions Other Distributions, Basic Statistics, Correlation and Covariance, T-Tests, Hypothesis Testing and ANOVA.

Unit III:**(10 Lectures)**

Exploratory Data Analysis: Data Visualization using R, Reading and getting data into R (External Data): Using CSV files, XML files, Web Data, JSON files, Databases, Excel files. Working with R Charts and Graphs: Histograms, Boxplots, Bar Charts, Line Graphs, Scatterplots, Pie Charts.

Unit IV:**(10 Lectures)**

Classification and Clustering Techniques: Overview of Classification, Decision Trees, Decision Tree Algorithms, Decision Tree in R, Bayes Theorem, Naïve Bayes Classifier, Smoothing, Naïve Bayes in R, Random Forests. Overview of Clustering, K-means, Use Cases, Overview of the Method, perform a K-means Analysis using R.

Unit V:**(09 Lectures)**

Regression Techniques: Linear Models, Simple Linear Regression, -Multiple Regression Generalized Linear Models, Logistic Regression, - Poisson Regression- other Generalized Linear Models-Survival Analysis, Nonlinear Models, Splines etc. and Case studies.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Understand the important keywords of Data Analytics and also learn how R Programming can play an important role in solving complex analytical problems.
2. Apply the knowledge of probability and statistics to test the significance of the data to be analyzed.
3. Identify the exploratory data analysis is an important step in the data analytics and also to work with different data formats.
4. Distinguish in between supervised and unsupervised machine learning algorithms for data modeling.
5. Analyze the data models to predict the future outcomes.

Textbooks:

1. An Introduction to R, Notes on R: A Programming Environment for Data Analysis and Graphics. W. N. Venables, D.M. Smith and the R Development Core Team. Version 3.0.1 (2013-05-16). URL: <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>
2. David Dietrich, Barry Heller and Beibei Yang, "Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data", EMC Education Services, Reprint 2015, Wiley, ISBN: 9788126556533.

References:

1. Big Data-Principles and best practices of scalable real-time data systems, R in Action, Rob Kabacoff, Manning Nathan Marz, James Warren, 2015, DreamTech Press. ISBN:9789351198062
2. The art of R programming: tour of statistical software design, Norman Matloff. ISBN-13: 978-1-59327-384-2.

JULIA PROGRAMMING

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PC	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Probability and Statistics, Data Analytics

Objectives: The Students will learn the following:

1. Understand the data science concepts and terminologies.
2. Interpret the data using basic statistics and data structures.
3. Explore data and analyze the data by using R functions and Packages.
4. Implementation of classification, clustering and feature selection methods with R.
5. Model the data to analyze and to predict using regression techniques.

Unit I**(10 Lectures)**

Introduction: Data Science, Big Data, Julia, Julia and R Packages, Datasets, Variable Names, Operators, Types, Data Structures, Control Flow and Functions.

Unit II**(08 Lectures)**

Working with Data: Dataframes, Categorical Data, Input/output Data Access, Useful Dataframe Functions, Split-Apply-Combine Strategy, Query.JL.

Unit III**(08 Lectures)**

Visualizing Data: Visualizing Univariate Data, Distributions, Visualizing Bivariate Data, Error Bars, Facets, Saving Plots, Gadfly.JL.

Unit IV**(12 Lectures)**

Supervised Learning: Introduction, Cross-Validation- K-Fold Cross-Validation, K-Nearest Neighbors Classification, Classification and Regression Trees, Bootstrap, Random Forests, Gradient Boosting.

Unit V**(12 Lectures)**

Unsupervised Learning: Introduction, Principal Components Analysis, Probabilistic Principal Components Analysis, K-Means algorithm, EM Algorithm for PPCA, R Interoperability- Interacting with R, Clustering and Data Reduction.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Understand the important keywords of Data Analytics and also learn how R Programming can play an important role in solving complex analytical problems.
2. Apply the knowledge of probability and statistics to test the significance of the data to be analyzed.
3. Identify the exploratory data analysis is an important step in the data analytics and also to work with different data formats.
4. Distinguish in between supervised and unsupervised machine learning algorithms for data modeling.
5. Analyze the data models to predict the future outcomes.

Textbooks:

1. Data Science with Julia, Paul D. Mc Nicholas and Peter A. Tait, CRC Press, Taylor and Francis Group, ISBN: 978-1-138-49998-0.

References:

1. Think Julia -How to think Like a Computer Scientist by Ben Lauwens and Allen B. Downey O'Reilly Media, ISBN: 978-1-492-04503-8
2. Learn Julia Language by Free Unaffiliated eBook created from Stack Overflow Contributors.

HUMAN COMPUTER INTERACTION (PROFESSIONAL ELECTIVE V)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-V	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites:**Objectives:** The Students will learn the following:

1. To apply models from cognitive psychology to predicting user performance in various human computer interaction
2. Tasks and recognize the limits of human performance as they apply to computer operation.
3. To provide the future user interface designer with concepts and strategies for computer science on user interface development projects.
4. Understand the social implications of technology and their ethical responsibilities as engineers in the design of technological systems.

Unit I**(08 Lectures)**

Introduction: Importance of user Interface, definition, importance of good design. Benefits of good design. A brief history of Screen design. The graphical user interface, Popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user, Interface popularity, characteristics, Principles of user interface.

Unit II**(10 Lectures)**

Design Process: Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions.

Screen Designing: Design goals, Screen planning and purpose, organizing screen elements, ordering of screen data and content, screen navigation and flow, Visually pleasing composition, amount of information, focus and emphasis, presentation information simply and meaningfully, information retrieval on web, statistical graphics, Technological consideration in interface design.

Unit III**(08 Lectures)**

Windows: New and Navigation schemes selection of window, selection of devices based and screenbased controls. Components, text and messages, Icons and increases, Multimedia, colors, uses problems, choosing colors.

Unit IV**(12 Lectures)**

Software Process: HCI in the software process, The software life cycle Usability engineering Iterative design and prototyping Design Focus: Prototyping in practice Design rationale Design rules Principles to support usability Standards Golden rules and heuristics HCI patterns Evaluation techniques, Goals of evaluation, Evaluation through expert analysis, Evaluation through user participation, Choosing an evaluation method. Universal design, Universal design principles Multi-modal interaction.

Unit V**(10 Lectures)**

Cognitive Models Goal and Task Hierarchies Design Focus: GOMS saves money Linguistic models The challenge of display based systems Physical and device models Cognitive architectures Ubiquitous computing and augmented realities Ubiquitous computing applications research Design Focus: Ambient Wood, augmenting the physical Virtual and augmented reality Design Focus: Shared experience Design Focus: Applications of augmented reality Information and data visualization Design Focus: Getting the size right.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Know evaluation and usability testing methods to interactive products to validate design decisions.
2. Design certain tools for blind or PH people.
3. Understand selection of devices based and screen based controls.
4. Learn importance of human characteristics human consideration.

Textbooks:

1. The Essential guide to user interface design, Wilbert O Galitz, Wiley DreamaTech.
2. Designing the user interface. 3rd Edition Ben Shneidermann, Pearson Education Asia.
3. Dix A., Finlay J., Abowd G. D. and Beale R. Human Computer Interaction, 3rd Edition, Pearson Education, 2005.

References:

1. Think Julia -How to think Like a Computer Scientist by Ben Lauwens and Allen B. Downey O'Reilly Media, ISBN: 978-1-492-04503-8.
2. Learn Julia Language by Free Unaffiliated eBook created from Stack Overflow Contributors.

INTRODUCTION TO ROBOTIC PROCESS AUTOMATION (PROFESSIONAL ELECTIVE V)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-V	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Machine Learning

Objectives: The Students will learn the following:

1. Explain the use of RPA platform in performing a task.
2. Apply sequence and control flows as per the requirements.
3. Analyse data manipulation concepts to solve real time problems.
4. Evaluate dynamic user interface explorer and handle events.
5. Analyse the scenario of handling the errors and exceptions to deploy a bot

Unit I

(11 Lectures)

Introduction to Robotic Process Automation: Scope and techniques of automation, Benefits of RPA, Components of RPA, RPA platforms, About UiPath. Record and Play: UiPath stack, Downloading and installing UiPath Studio, Learning UiPath Studio, Task recorder, Step-by-step examples using the recorder.

Unit II

(09 Lectures)

Sequence & Control Flow: Sequence, Flowchart, and Control Flow, Sequencing the workflow, Activities, Control flow, various types of loops, and decision making, Step-by-step example using Sequence and Flowchart, Step-by-step example using Sequence and Control flow.

Unit III

(10 Lectures)

Data Manipulation: Variables and scope, Collections, Arguments-purpose and use, Data table usage with examples, Clipboard management, File operation with step-by-step example, CSV/Excel to data table and vice versa.

Unit IV

(10 Lectures)

Working with UI Explorer: Handling events: Element triggering events, image triggering events, system triggering events, PDF Extraction, Revisit Recorder: Basic recording, Desktop recording, web recording, Screen Scraping, Automation Techniques: Incoming Email automation, Sending Email automation, Workbook and Excel automation (Read/Write).

Unit V

(09 Lectures)

Error and Exception Handling: Exception handling, Common exceptions and ways to handle them, Logging and taking screenshots, debugging techniques, Collecting crash dumps, Error reporting. Deploying and Maintaining the Bot, Publishing using publish utility, Overview of Orchestration Server, Using Orchestration Server to control bots, Using Orchestration Server to deploy bots.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Explain the use of RPA platform in performing a task.
2. Apply sequence and control flows as per the requirements.
3. Analyse data manipulation concepts to solve real time problems.
4. Evaluate dynamic user interface explorer and handle events.
5. Analyse the scenario of handling the errors and exceptions to deploy a bot.

Textbook:

1. Learning Robotic Process Automation, Alok Mani Tripathi, Packt Publishing, 2018
2. Introduction to Robotic Process Automation: a Primer, Institute of Robotic Process Automation, Frank Casale, Rebecca Dilla, Heidi Jaynes, Lauren Livingston, 2015.

References:

1. Richard Murdoch, Robotic Process Automation: Guide to Building Software Robots, Automate Repetitive Tasks & Become an RPA Consultant, Independently Published, 2018.
2. Robotic Process Automation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation, Srikanth Merianda, Create Space Independent Publishing Platform, 2018.

SPEECH AND AUDIO PROCESSING
(PROFESSIONAL ELECTIVE V)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-V	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Machine Learning, Natural Language Processing, Probability And Statistics

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E

Unit I

(11 Lectures)

Fundamentals of Speech: Human Speech Production Mechanism, LTI Model For Speech Recognition, Nature of Speech Signal, Linear-Time Varying Model, Phonetics, Types of Speech, Voiced and Unvoiced Decision Making.

Unit II

(09 Lectures)

Parameters of Speech - Pitch and Formants: Pitch Frequency, Parallel Processing Approach for Calculation of Pitch Frequency, Pitch Period Measurement Using Spectral Domain, Cepstral Domain, Formants and Their Relation With LPC.

Unit III

(12 Lectures)

Spectral Parameters of Speech: Homomorphic Processing, Cepstral Analysis of Speech - Cepstral Coefficients, Auditory System as a Filter Bank, Mel Frequency Cepstral Coefficients (MFCCs).

Perceptual Linear Prediction (PLP): Introduction to Perceptual Linear Prediction (PLP), Relative Spectral Perceptual Linear Prediction (Rasta-PLP), Short-Time Spectral Analysis of Speech, Wavelet Transform Analysis of Speech.

Unit IV

(10 Lectures)

Speech Quantization and Coding: Uniform and Non-Uniform Quantizers and Coder, Companded Quantizer, Uniform Quantization of Non-uniform Sources, Waveform Coding of Speech.

Unit V

(09 Lectures)

Waveform Coding Techniques: Parametric Speech Coding Techniques, Sinusoidal Speech Coding Techniques, Mixed Excitation Linear Prediction Coder, Multi-Mode Speech Coding (Hybrid Coder), Transform Domain Coding of Speech.

Outcomes: Upon the successful completion of the course, the student will be able:

1. A
2. B
3. C
4. D
5. E.

Textbook:

1. Speech and Audio Processing, Dr. Shaila D. Apte, Wiley Publications.

References:

1. A
2. B

**SOCIAL MEDIA ANALYTICS
(PROFESSIONAL ELECTIVE V)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-V	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Data Warehousing and Data Mining

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E

Unit I

(11 Lectures)

Data Identification: Attributes of Data – Structure, Language, Region, Type of Content, Venue, Time, Ownership of Data, Casting a Net, Regular Expression.

Unit II

(09 Lectures)

Comments Interested In: Right Subset of People – Employment, Sentiment, Location, Language, Gender, Profession, Popularity, Role, Predictive Vs Descriptive Analytics.

Unit III

(12 Lectures)

Social Data: Structured Vs Unstructured Data, Big Data, Identifying Data in Social Media Outlets – Professional Networking Sites, Social Sites, Information Sharing Sites, Microblogging Sites.

Unit IV

(10 Lectures)

Data Analysis: Four Dimensions of Analysis Taxonomy, Depth of Analysis, Machine Capacity, Domain of Analysis, Velocity of Data

Unit V

(09 Lectures)

Discovering Themes, Topics, Trends: Validating the Hypothesis, Discovering Themes and Topics, Iterative Methods.

Outcomes: Upon the successful completion of the course, the student will be able:

1. A
2. B
3. C
4. D
5. E

Textbook:

1. Social Media Analytics – Techniques and Insights for Extracting Business Value out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press.

References:

1. A
2. B
3. C
4. D
5. E

**SUPPLY CHAIN ANALYTICS
(PROFESSIONAL ELECTIVE V)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-V	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Data Warehousing and Data Mining

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E

Unit I **(11 Lectures)**

Introduction: Defining the Supply Chain Analytics (SCA), Supply Chain Management, Importance of Analytics in Supply Chain, Operations Management with Supply Chain Concepts with SC Analytics, Importance of SCA in the flows involving Material, Money, Information, and Ownership.

Unit II **(09 Lectures)**

Supply Chain Analytics: Key Issues, Components, Descriptive Analytics in a Supply Chain, Decision Domains in SCA.

Unit III **(12 Lectures)**

Foundation of Business Analytics (BA): Introduction to Modelling, Approaches for Optimization and Simulation, Modelling Software, Supply Chain (SC) Decisions that require Mathematical Modelling, Data and its Role in Analytics, Analytics of a Transportation Problem in a Supply Chain Managerial Implication of Results of Analytics.

Unit IV **(10 Lectures)**

Foundation of Prescriptive Analytics in Network Planning in a Supply Chain: Network Planning in a Supply Chain, Importance of Network Planning, Design of Logistics Network using Heuristics/optimization, Concept of 3PL/4PL in a Supply Chain, Case Study: GATI.

Unit V **(09 Lectures)**

Supply Chain Management: Modelling Coordination Decisions, Performance Management, IT Enablement of Supply Chains, Role of ICT in Supply Chains.

Outcomes: Upon the successful completion of the course, the student will be able:

1. A
2. B
3. C
4. D
5. E.

Textbook:

1. Supply Chain Management, Sunil Chopra, Peter Meindl, Pearson Education.

References:

1. Modeling the Supply Chain, Jeremy F. Shapiro, Duxbury Thomson Learning
2. Business Analytics - A Practitioner's Guide, Rahul Saxena, Anand Srinivasan, Springer.

**INFORMATION RETRIEVAL SYSTEMS
(PROFESSIONAL ELECTIVE VI)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-VI	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Data Warehousing and Data Mining, Probability and Statistics

Objectives: The Students will learn the following:

1. Identify the theoretical basis behind the standard models of Information Retrieval
2. Interpret the difficulty of representing and retrieving documents, images, speech, etc.,
3. Implement, run and test a standard Information Retrieval system,
4. Explain the standard methods for Web indexing and retrieval,
5. Infer how techniques from natural language processing, artificial intelligence, human-computer interaction and visualization integrate with Information Retrieval

Unit I

(11 Lectures)

Retrieval Strategies: Vector space model, Probabilistic Retrieval Strategies: Simple term weights, Non-Binary Independence Model, Language models.

Unit II

(09 Lectures)

Retrieval Utilities: Relevance feedback, Clustering, N-Grams, Regression Analysis, Thesauri, Evaluation in information retrieval.

Unit III

(12 Lectures)

Retrieval Utilities: Semantic Networks, Parsing.

Cross-Language Information Retrieval: Introduction, Crossing the language Barrier.

Unit IV

(10 Lectures)

Efficiency: Inverted Index, Query Processing, Signature files, Duplicate Document detection, Matrix decompositions and latent semantic indexing.

Unit V

(09 Lectures)

Integrating Structure Data and text: A historical Progression, Information Retrieval as a relational Application, Semi-Structured search using a Relational Schema.

Distributed Information Retrieval: A theoretical Model of Distributed Retrieval, Web Search.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Apply IR principles to locate relevant information large collections of data
2. Analyze performance of retrieval systems when dealing with unmanaged data sources
3. Implement techniques for the preprocessing needed for information retrieval systems.
4. Illustrate the web indexing and retrieval methods.
5. Implement retrieval systems for web search tasks.

Textbook:

1. Information Retrieval: Algorithms and Heuristics, David A. Grossman, Ophir Frieder, Springer, 2nd Edition (Distributed by Universities Press), 2004.

References:

1. Information Storage and Retrieval Systems, Gerald J Kowalski, Mark T Maybury, Springer, 2000.
2. Information Storage & Retrieval, Robert Korfage – John Wiley & Sons.
3. Introduction to Information Retrieval, Christopher D. Manning, Prabhakar Raghavan, Cambridge University Press, 2008.
1. Souren Chakrabarti, Miming the Web: Discovering Knowledge from Hypertext data, Morgan-Kaufmann Publishers, 2002.

**INTRODUCTION TO MEDICAL IMAGE ANALYSIS
(PROFESSIONAL ELECTIVE VI)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-VI	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Nil

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E

Unit I

(11 Lectures)

Introduction: Features of Image, Digital Image, Image Resolution and Aspect Ratio, Components of Digital Image Processing, Sampling and Quantization, Vision Fundamentals, CAD System.

Unit II

(09 Lectures)

Biomedical Image Processing: Modalities of Medical Imaging, Issues with Medical Images, Image Enhancement, Radiography, Position Emission Tomography, Computed Tomography Angiography, Echocardiogram.

Unit III

(10 Lectures)

Noise Reduction Filters for Medical Images: Sources of Noise, Filters Used for Noise Reduction, Spatial Domain Filters, Frequency Domain Filters.

Unit IV

(10 Lectures)

Feature Extraction and Statistical Measurement: Selection of Features, Shape Related Features, Fourier Descriptors, Texture Analysis, Breast Tissue Detection, Analysis of Tissue Structures..

Unit V

(09 Lectures)

Medical Image Restoration: Image Restoration, Degradation Model, Estimation of Degradation Function, Blur Model, Medical Image Restoration, Blur Identification, Super-Resolution Method, Applications of Image Restoration.

Outcomes: Upon the successful completion of the course, the student will be able:

1. A
2. B
3. C
4. D
5. E.

Textbook:

1. Medical Image Processing Concepts and Applications, G R Sinha, Bhagwati Charan Patel, PHI Learning.

References:

1. A
2. B
3. C
4. D
5. E

**MACHINE TRANSLATION
(PROFESSIONAL ELECTIVE VI)**

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-VI	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Natural Language Processing

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E

Unit I

(11 Lectures)

Introduction: Data-Driven Machine Translation (MT), MT Approaches – Ascending Transfer and Descending Transfer, Syntactic Divergence, Lexical-Semantic Divergence, Paradigms of Machine Translation, MT Evaluation.

Unit II

(09 Lectures)

Learning Bilingual Word Mappings: Combinatorial Argument, One-to-One Alignment, Heuristics-Based Computation, Iterative Computation, Derivation of Alignment Probabilities.

Unit III

(10 Lectures)

Model of Alignment: Influencing Factors, Problem of Summation Over Product in IBM Model 1, EM for Computing, Alignment in New Input Sentence Pair.

Phrase-Based Machine Translation: Need for Phrase Alignment, Phrase Alignment Techniques – Two-Way Alignments, Summarization, Principles of Phrase Construction.

Unit IV

(10 Lectures)

Rule-Based Machine Translation: Interlingua, Transfer, Universal Networking Language, UNL Expressions as Binary Predicates, Interlingua and Word Knowledge.

Unit V

(09 Lectures)

Medical Image Restoration: Image Restoration, Degradation Model, Estimation of Degradation Function, Blur Model, Medical Image Restoration, Blur Identification, Super-Resolution Method, Applications of Image Restoration.

Outcomes: Upon the successful completion of the course, the student will be able:

1. A
2. B
3. C
4. D
5. E.

Textbook:

1. Machine Translation, Pushpak Bhattacharya, CRC Press.

References:

1. A
2. B
3. C
4. D
5. E

NOSQL DATABASES (PROFESSIONAL ELECTIVE VI)

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-VI	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Database Management Systems

Objectives: The Students will learn the following:

1. To understand the emergence of NoSQL Databases, its application and Integration with other Databases.
2. Provide comparison of different NoSQL Databases with RDBMS.
3. To understand the features of Document oriented NoSQL Databases.
4. Provide understanding of Column-Family Data store features and Content Management Systems.
5. To understand the consistency and Query Features of Key-Value and Graph NoSQL Databases

Unit I

(11 Lectures)

Overview and History of NoSQL Databases: Definition of the Four Types of NoSQL Database, The Value of Relational Databases, Getting at Persistent Data, Concurrency, Integration, Impedance Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL, Key Points.

Unit II

(09 Lectures)

Aggregate Data Models and Distribution Models: Comparison of relational databases to new NoSQL stores, MongoDB, Cassandra, HBASE, Neo4j use and deployment, Application, RDBMS approach, Challenges NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Aggregate-Oriented Databases. Replication and sharding, Map Reduce on databases. Distribution Models, Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication.

Unit III

(10 Lectures)

Document oriented NoSQL Database: NoSQL Key/Value databases using MongoDB, Document Databases, Document oriented Database Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Web Analytics or Real-Time Analytics, E-Commerce Applications, Complex Transactions Spanning Different Operations, Queries against Varying Aggregate Structure.

Unit IV

(10 Lectures)

Column- oriented NoSQL Database: Column- oriented NoSQL databases using Apache HBASE, Column oriented NoSQL databases using Apache Cassandra, Architecture of HBASE, Column-Family Data Store Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Counters, Expiring Usage.

Unit V

(09 Lectures)

NoSQL Key-Value Database and Graph NoSQL Database: NoSQL Key/Value databases using Riak, Key Value Databases, Key-Value Store, Key-Value Store Features, Consistency, Transactions, Query Features, Structure of Data, Scaling, Suitable Use Cases, Storing Session Information, User Profiles, Preferences, Graph NoSQL databases using Neo4, NoSQL database development tools and programming languages, Graph Databases, Graph Database. Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Outline the different types of NoSQL Databases and the integration of NoSQL Databases.
2. Compare and contrast RDBMS with different NoSQL Databases.
3. Demonstrate the detailed architecture and performance tune of Document-oriented NoSQL Databases.
4. Analyze performance tune of Key-Value Pair NoSQL Databases.
5. Apply NoSQL development tools on different types of NoSQL Databases.

Textbook:

1. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Sadalage, P. & Fowler, Wiley Publications, 1st Edition, 2019.

References:

1. Getting Started with NoSQL, Your guide to the world and technology of NoSQL, Gaurav Vaish, Packt Publishing, 1st Edition, March 2013.
2. NoSQL Databases, Professional, Christof Strauch

INDUSTRIAL AND MEDICAL IOT

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV-I		PE-VI	L	T	P	C	CIE	SEE	Total
				3	-	-	3	40	60	100

Pre-requisites: Database Management Systems

Objectives: The Students will learn the following:

1. A
2. B
3. C
4. D
5. E

Unit I**(11 Lectures)**

Introduction to Industrial IoT: Introduction to IOT, Definition of Industrial IoT (IIOT), Use cases of IIoT, IOT vs. IIOT, IoT Analytics and AI. Industrial Process and Devices: Technical requirements, Industrial processes and types, CIM pyramid Architecture.

Unit II**(09 Lectures)**

Industrial Data Flow and Devices: Technical requirements, IIoT Data flow in the factory, Measurements and the Actuator Chain, IIoT Devices- Sensors, Controllers, PLCs, DCS, Industrial Protocols.

Unit III**(10 Lectures)**

Implementing Industrial IoT Data Flow: Discovering OPC, OPC classic, OPC UA, Understanding IIoT edge, features of the edge, The Edge Architecture and Implementation, IIoT data sources and data gathering, Edge Deployment and Data flow scenarios.

Unit IV**(10 Lectures)**

IoMT Introduction: IoMT and its working, Tracking assets and resources, Internet of things in hospitals, collection and integration of clinical data, Major benefits of IoT in healthcare, Disadvantages of IoT in healthcare.

Unit V**(09 Lectures)**

Healthcare Technologies: Home Monitoring System for Aged Care, Smart Medicinal Packages for Medication Adherence, Smart Drug Delivery System for Automated Drug Dispensation, Connected Rural Healthcare Consultation, Population and Environment Monitoring of Infectious Diseases.

Outcomes: Upon the successful completion of the course, the student will be able:

1. A
2. B
3. C
4. D
5. E

Textbook:

1. Hands-on Industrial Internet of Things: Create a Powerful Industrial IoT Infrastructure Using Industry 4.0, Veneri, Giacomo, and Antonio Capasso, 1st Edition, Packt Publishing Ltd, 2018.
2. Internet of Things and advanced application in healthcare, Reis, Catarina I., Marisa da Silva Maximiano, 1st Edition, IGI Global.

References:

1. Industry 4.0: The Industrial Internet of Things, Alasdair Gilchrist., 1st Edition, Apress, 2017
2. Medical Big Data and Internet of Medical Things: Advances, Challenges and Applications, Aboul Ella Hassanien, Nilanjan Dey, SureekaBoara, 1st Edition, CRC Press, 2019.

DATA VISUALIZATION TECHNIQUES LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV –I		PC	L	T	P	C	CIE	SEE	Total
				-	-	3	1.5	40	60	100

Pre-requisites:

Objectives: The Students will learn the following:

1. Understand the practical application of data visualization tools and techniques for effective data representation and analysis.
2. Gain hands-on experience in using various data visualization software and libraries to create different types of visualizations.
3. Develop skills in designing and customizing data visualizations to meet specific requirements and enhance data communication.
4. Analyze and interpret data visualizations to draw meaningful insights and support data-driven decision-making.
5. Explore advanced data visualization methods and techniques, including interactive and dynamic visualizations.

List of Programs:

Week 1: Exploring data visualization software and basic charts.

Week 2: Creating effective visualizations using different encodings.

Week 3: Exploring and visualizing datasets using EDA techniques.

Week 4: Creating interactive maps and visualizing geospatial data.

Week 5: Creating network visualizations using graph visualization libraries.

Week 6: Creating visualizations for time series and temporal data.

Week 7: Visualizing high-dimensional data using dimensionality reduction.

Week 8: Visualizing text data and extracting insights.

Week 9: Creating interactive and web-based visualizations.

Week 10: Designing and building interactive data dashboards.

Week 11: Visualizing machine learning models and interpretability techniques.

Week 12: Developing and presenting a data visualization project.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Demonstrate proficiency in using data visualization tools to create accurate and informative visual representations of data.
2. Design and develop a variety of data visualizations, showcasing data in a visually compelling and intuitive manner.
3. Analyze and critique data visualizations for effectiveness, accuracy, and adherence to design principles.
4. Interpret data visualizations to identify patterns, trends, and outliers, and present findings coherently.
5. Generate interactive and dynamic data visualizations that allow for exploration and interactive analysis of data.

Textbooks:

1. The Visual Display of Quantitative Information, Edward R. Tufte. Publisher: Graphics Press; 2nd Edition, 2001.
2. Storytelling with Data: A Data Visualization Guide for Business Professionals, Cole Nussbaumer Knaflic. Wiley, 1st Edition, 2015.
3. "Information Visualization: Perception for Design" by Colin Ware. Publisher: Morgan Kaufmann; 3rd Edition (2012).
4. "Data Visualization: A Practical Introduction" by Kieran Healy. Publisher: Princeton University Press; 1st Edition (2018).
5. "Interactive Data Visualization for the Web: An Introduction to Designing with D3" by Scott Murray. Publisher: O'Reilly Media; 2nd Edition (2017).

JULIA PROGRAMMING LAB

Regulation	Year/ Sem	Course Code	Category	Hours / Week			Credits	Maximum Marks		
R22	IV –I		PC	L	T	P	C	CIE	SEE	Total
				-	-	3	1.5	40	60	100

Pre-requisites:

Objectives: The Students will learn the following:

1. How to program in R and how to use R for effective data analysis.
2. Install and configure software necessary for a statistical programming environment.
3. Carry out basic data pre-processing and wrangling in R-Studio and to explore practical issues in statistical computing.
4. Read in data into the R environment from different sources
5. Implement supervised/unsupervised machine learning techniques.

Recommended Systems/Software Requirements:

Download the current stable version of Julia (1.4.0) corresponding to your operating system, and install it.

URL: <https://julialang.org/downloads>

List of Programs:

Week 1: Overview of Julia Platform, working with data types and variables.

Week 2: Control flow and functions.

Week 3: Data Structures (Dataframes, Arrays and Dictionaries)

Week 4: Working with different datasets, Input / Output file access, Dataframe functions

Week 5: Visualizing Univariate and Bivariate Data Distributions

Week 6: Visualizing Error Bars, Facets and Saving Plots

Week 7: K-Fold Cross Validation and K- Nearest Neighbour Classification

Week 8: Working with Classification and Regression Trees

Week 9: Working with Random Forests and Gradient Boosting algorithms

Week 10: K- Means Clustering

Week 11: Case Study- **Probabilistic Principal Components Analyzers**

- COFFEE DATA
- FOOD DATA

Week 12: Case Study- **Clustering and Data Reduction**

- COFFEE DATA
- FOOD DATA.

Outcomes: Upon the successful completion of the course, the student will be able:

1. Infer the R Programming language conventions for data analysis.
2. Interpret the functioning of datatypes, variables and control structures on R Platform.
3. Assess the significance of data/ datasets by applying the statistical tests using R.
4. Evaluate the performance of machine learning models.
5. Create and edit visualizations with R.

Textbook:

1. Data Science with Julia by Paul D. Mc Nicholas and Peter A. Tait, CRC Press, Taylor and Francis Group, ISBN: 978-1-138-49998-0.
2. Learn Julia Language by Free Unaffiliated eBook created from Stack Overflow Contributors